04 Seaborn

Plan for this lecture

1. Seaborn Introduction
2. Comparing similar Matplotlib and Seaborn plots
3. Seaborn's unique plotting functions
4. Exercises

Introduction to Seaborn

seaborn

- Seaborn is a Python visualisation library based on Matplotlib.
- Seaborn provides functions to create complex statistical plots with minimal code. Unlike Matplotlib, where all labels, colour, dimensions, size etc. need to be set.
- Seaborn has functions such as scatterplot(), barplot() and heatmap() which encapsulate common plotting patterns.
- More documentation available at: https://seaborn.pydata.org

Imports

In [1…

```
import seaborn as sns
```

In [1…

```
sns
```

Out[1…

```
<module 'seaborn' from '/Users/nick/Library/Python/3.9/lib/python/site-packages/seaborn/__init__.py'>
```

In [1…

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
```

`set_style()`

In […

```
sns.set_style('whitegrid')
```

In […

```
sns.set_style('darkgrid')
```

In […

```
sns.set_style()
```

`load_dataset()`

- returns a Pandas `DataFrame`

In [1…

```
titanic = sns.load_dataset('titanic')
titanic.head()
```

Out[1…

	survived	pclass	sex	age	sibsp	parch	fare	embarked	class	who	adult_male	deck	embark_town	alive	alone
0	0	3	male	22.0	1	0	7.2500	S	Third	man	True	NaN	Southampton	no	False
1	1	1	female	38.0	1	0	71.2833	C	First	woman	False	C	Cherbourg	yes	False
2	1	3	female	26.0	0	0	7.9250	S	Third	woman	False	NaN	Southampton	yes	True
3	1	1	female	35.0	1	0	53.1000	S	First	woman	False	C	Southampton	yes	False
4	0	3	male	35.0	0	0	8.0500	S	Third	man	True	NaN	Southampton	no	True

In [1…

```
type(titanic)
```

Out[1…

```
pandas.core.frame.DataFrame
```

In [1…

```
titanic.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 891 entries, 0 to 890
Data columns (total 15 columns):
 #   Column       Non-Null Count  Dtype   
---  ------       --------------  -----   
 0   survived     891 non-null    int64   
 1   pclass       891 non-null    int64   
 2   sex          891 non-null    object  
 3   age          714 non-null    float64 
 4   sibsp        891 non-null    int64   
 5   parch        891 non-null    int64   
 6   fare         891 non-null    float64 
 7   embarked     889 non-null    object  
 8   class        891 non-null    category
 9   who          891 non-null    object  
 10  adult_male   891 non-null    bool    
 11  deck         203 non-null    category
 12  embark_town  889 non-null    object  
 13  alive        891 non-null    object  
 14  alone        891 non-null    bool    
dtypes: bool(2), category(2), float64(2), int64(4), object(5)
memory usage: 80.7+ KB
```

In [2…

```
iris = sns.load_dataset('iris')
iris.head()
```

Out[2…

	sepal_length	sepal_width	petal_length	petal_width	species
0	5.1	3.5	1.4	0.2	setosa
1	4.9	3.0	1.4	0.2	setosa
2	4.7	3.2	1.3	0.2	setosa
3	4.6	3.1	1.5	0.2	setosa
4	5.0	3.6	1.4	0.2	setosa

In [2…

```
iris.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 150 entries, 0 to 149
Data columns (total 5 columns):
 #   Column        Non-Null Count  Dtype  
---  ------        --------------  -----  
 0   sepal_length  150 non-null    float64
 1   sepal_width   150 non-null    float64
 2   petal_length  150 non-null    float64
 3   petal_width   150 non-null    float64
 4   species       150 non-null    object 
dtypes: float64(4), object(1)
memory usage: 6.0+ KB
```

`pd.read_csv()` for local csv files

In [2…

```
df = pd.read_csv('AMZN-quote-history.csv')
df.head()
```

Out[2…

	Date	Open	High	Low	Close	Adj Close	Volume
0	2021-01-19	3107.000000	3145.000000	3096.000000	3120.760010	3120.760010	3305100
1	2021-01-20	3181.989990	3279.800049	3175.000000	3263.379883	3263.379883	5309800
2	2021-01-21	3293.000000	3348.550049	3289.570068	3306.989990	3306.989990	4936100
3	2021-01-22	3304.310059	3321.909912	3283.159912	3292.229980	3292.229980	2821900
4	2021-01-25	3328.500000	3363.889893	3243.149902	3294.000000	3294.000000	3749800

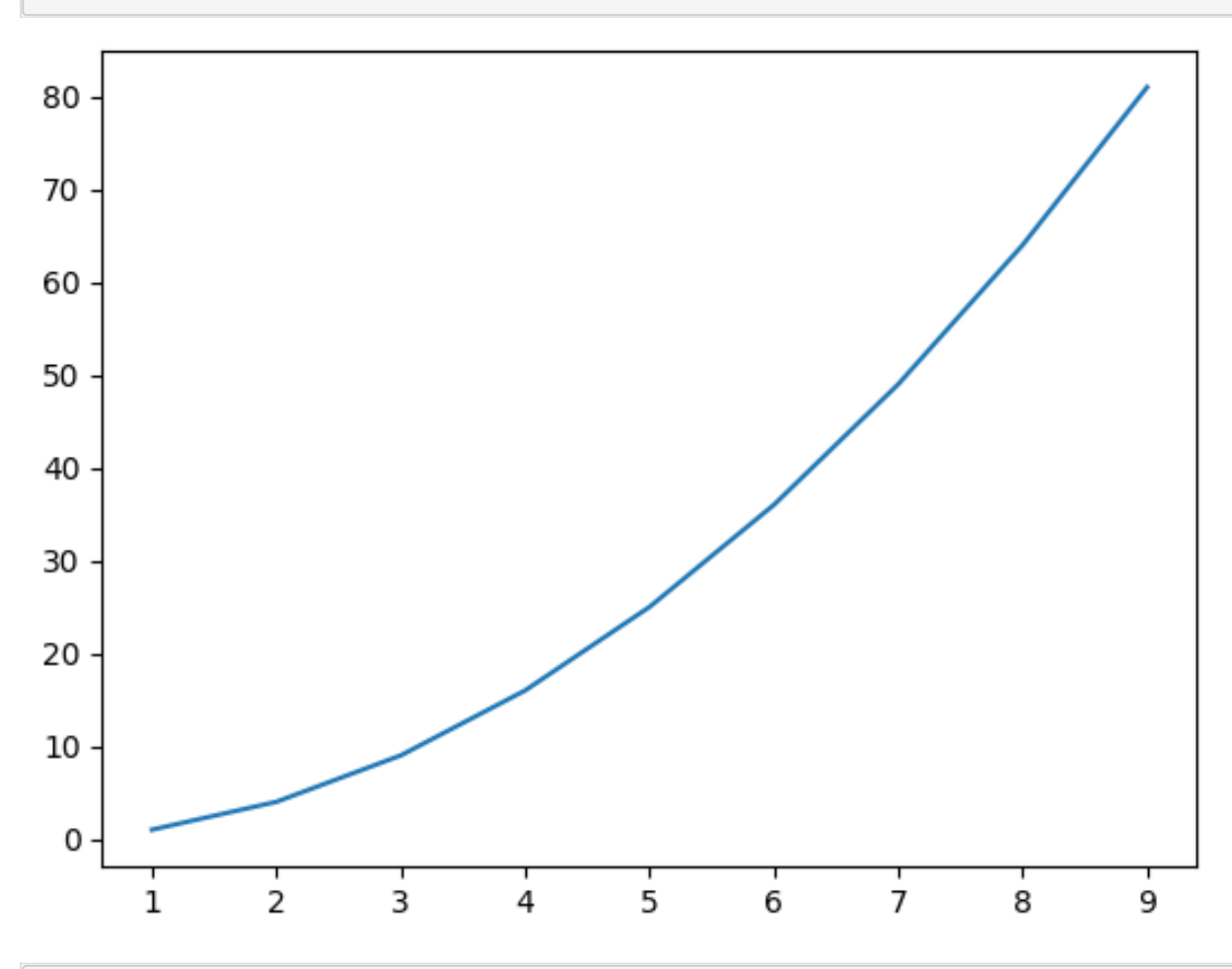
Comparison of `plt` and `sns` plots

- `plt.scatter()` vs `sns.scatter()` & `sns.regplot()`
- `plt.boxplot()` vs `sns.boxplot()`
- `plt.bar()` & `plt.hist()` vs `sns.barplot()` & `sns.catplot()`

Reminder: `plt's plot()`

In [2…

```
x = np.arange(1, 10, 1)
y = x**2
plt.plot(x, y)
plt.show()
```



In [2…

```
x
```

Out[2…

```
array([1, 2, 3, 4, 5, 6, 7, 8, 9])
```

In [2…

```
y
```

Out[2…

```
array([ 1,  4,  9, 16, 25, 36, 49, 64, 81])
```

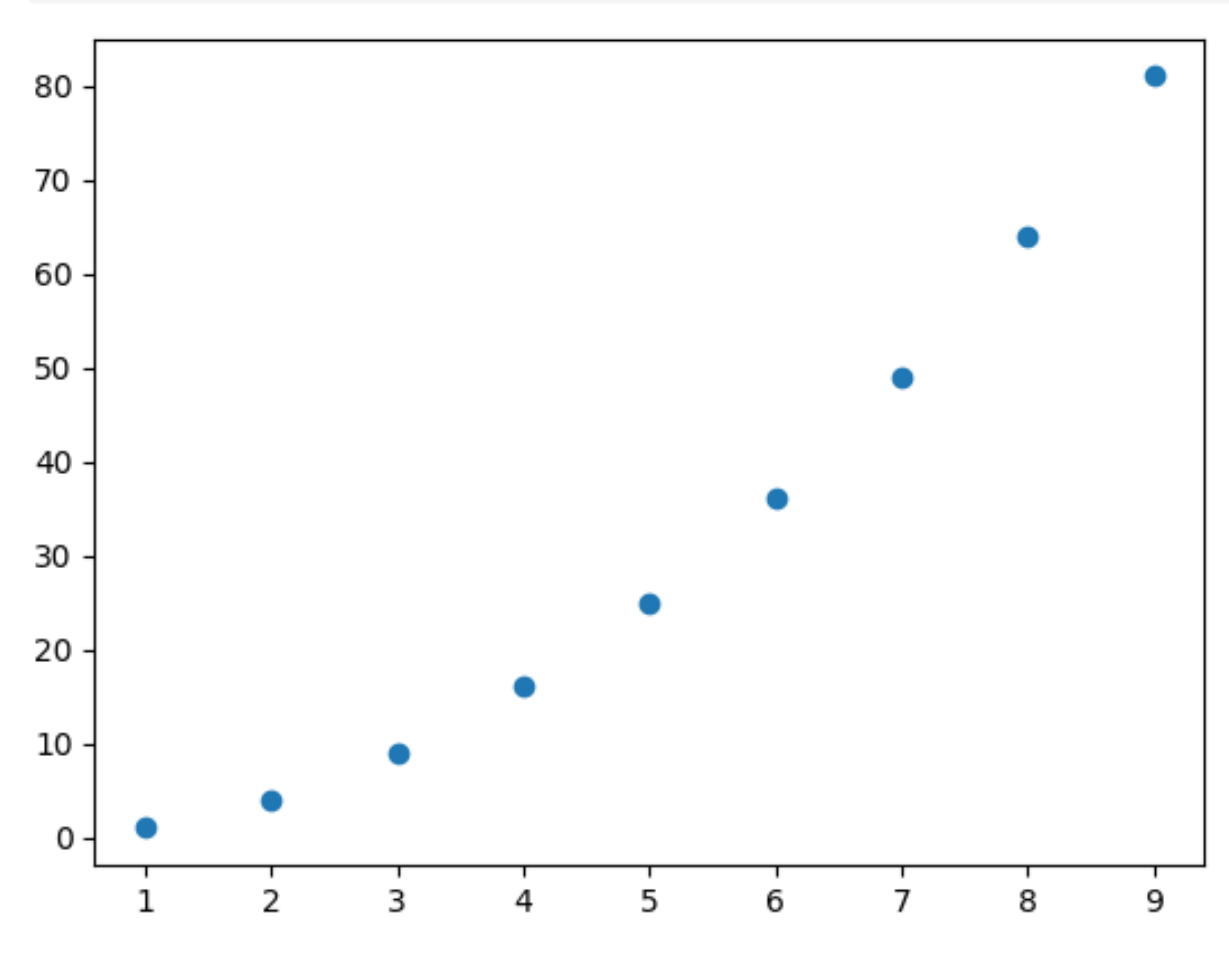
`plt's scatter()` plot

In [2…

```
x = np.arange(1, 10, 1)
y = x**2
plt.scatter(x, y)
```

Out[2…

```
<matplotlib.collections.PathCollection at 0x150081460>
```



`sns.scatter()`

- We can't use exactly the same arguments as we did for plt...

In [3…

```
x = np.arange(1, 10, 1)
y = x**2
sns.scatterplot(x, y)
```

```
---------------------------------------------------------------------------
TypeError                                 Traceback (most recent call last)
Cell In [30], line 3
      1 x = np.arange(1, 10, 1)
      2 y = x**2
----> 3 sns.scatterplot(x, y)

TypeError: scatterplot() takes from 0 to 1 positional arguments but 2 were given
```

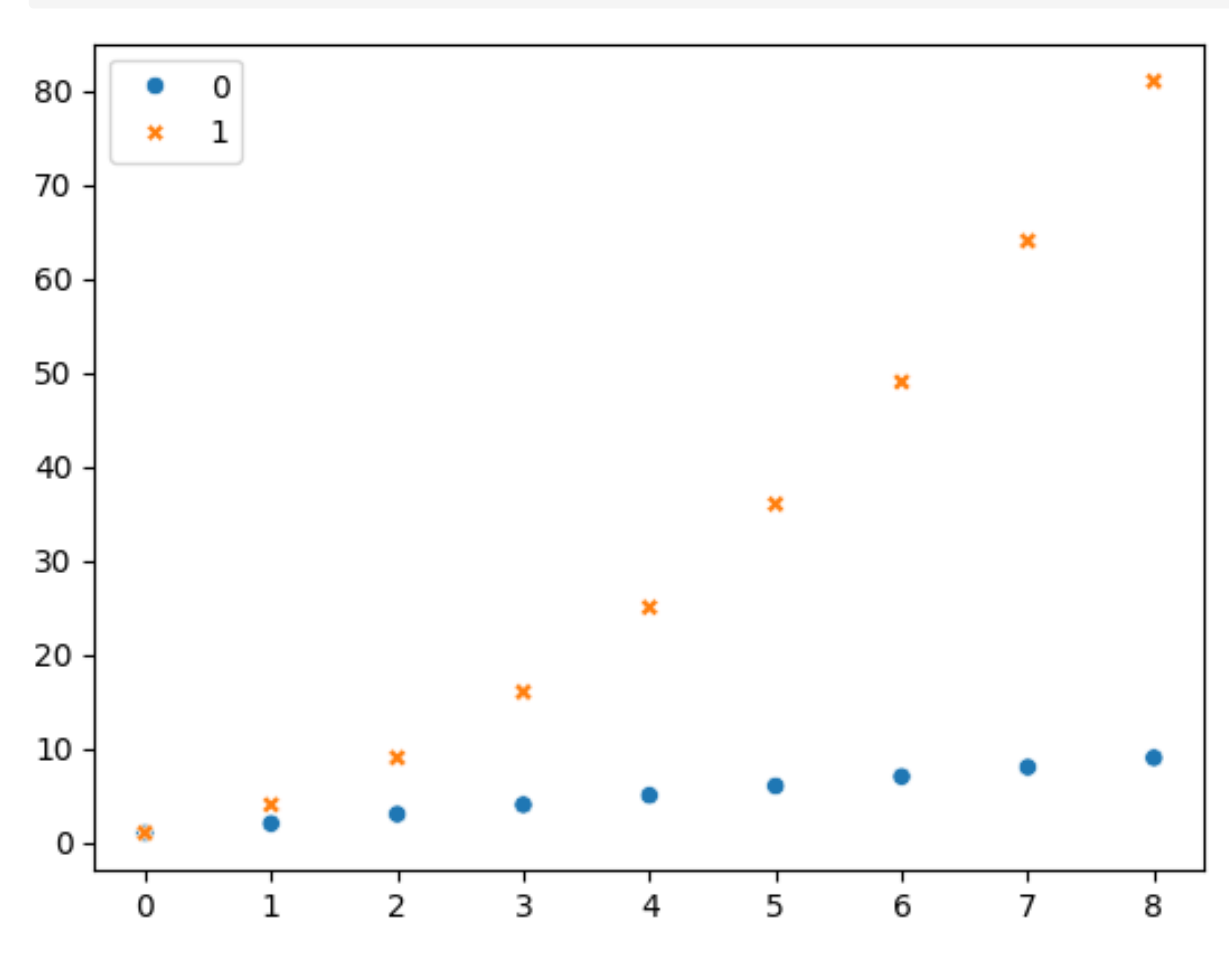
We could resolve this by bundling them as as a `tuple` : `(x, y)`

In [3…

```
x = np.arange(1, 10, 1)
y = x**2
sns.scatterplot(data = (x, y))
```

Out[3…

```
<Axes: >
```



But notice that this draws two sets of observations... x vs x, and y vs x !

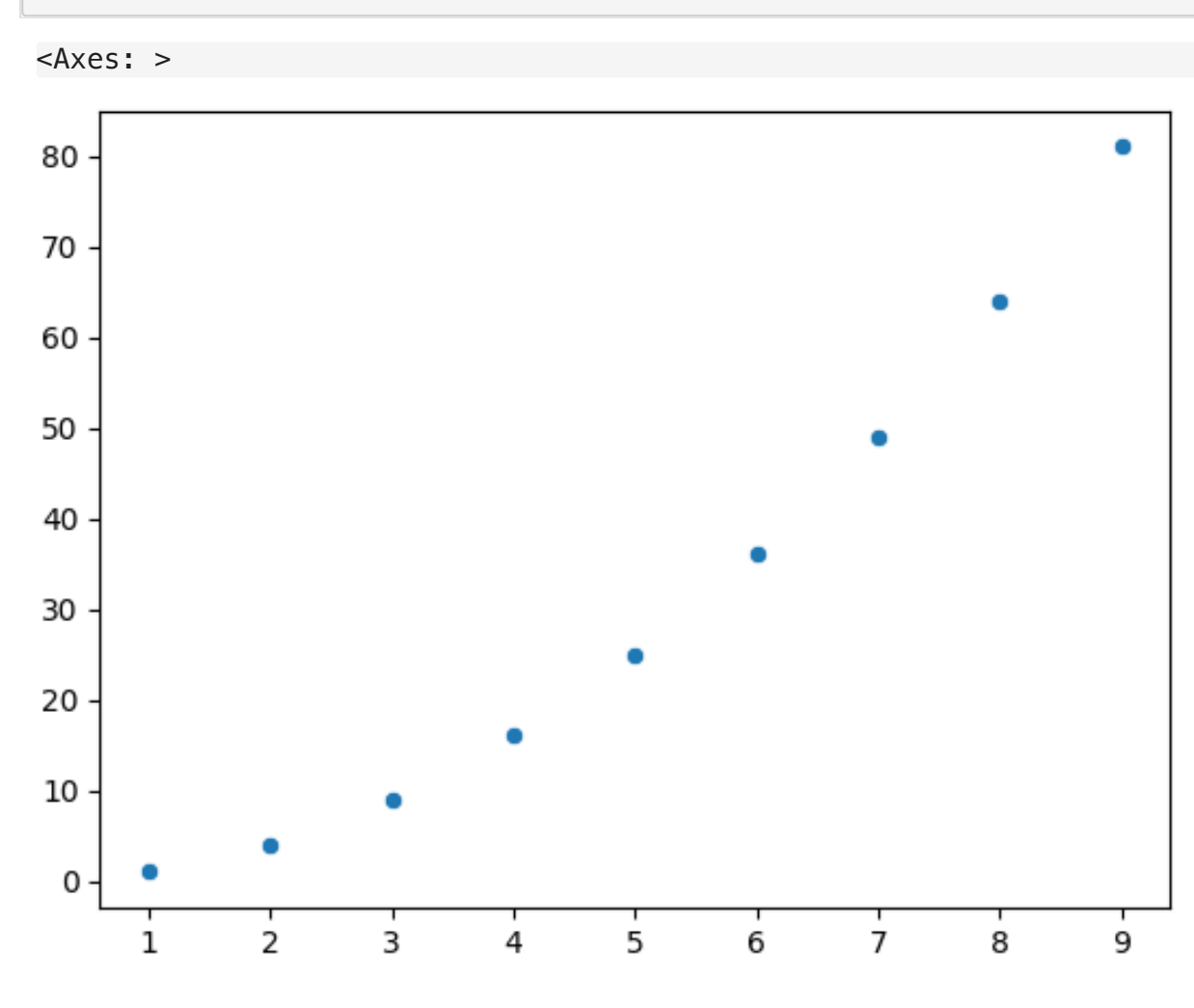
If we want one dataset: x vs y, we can use the `x` and `y` arguments:

In [3…

```
x = np.arange(1, 10, 1)
y = x**2
sns.scatterplot(x=x, y=y)
```

Out[3…

```
<Axes: >
```



`sns.regplot()`

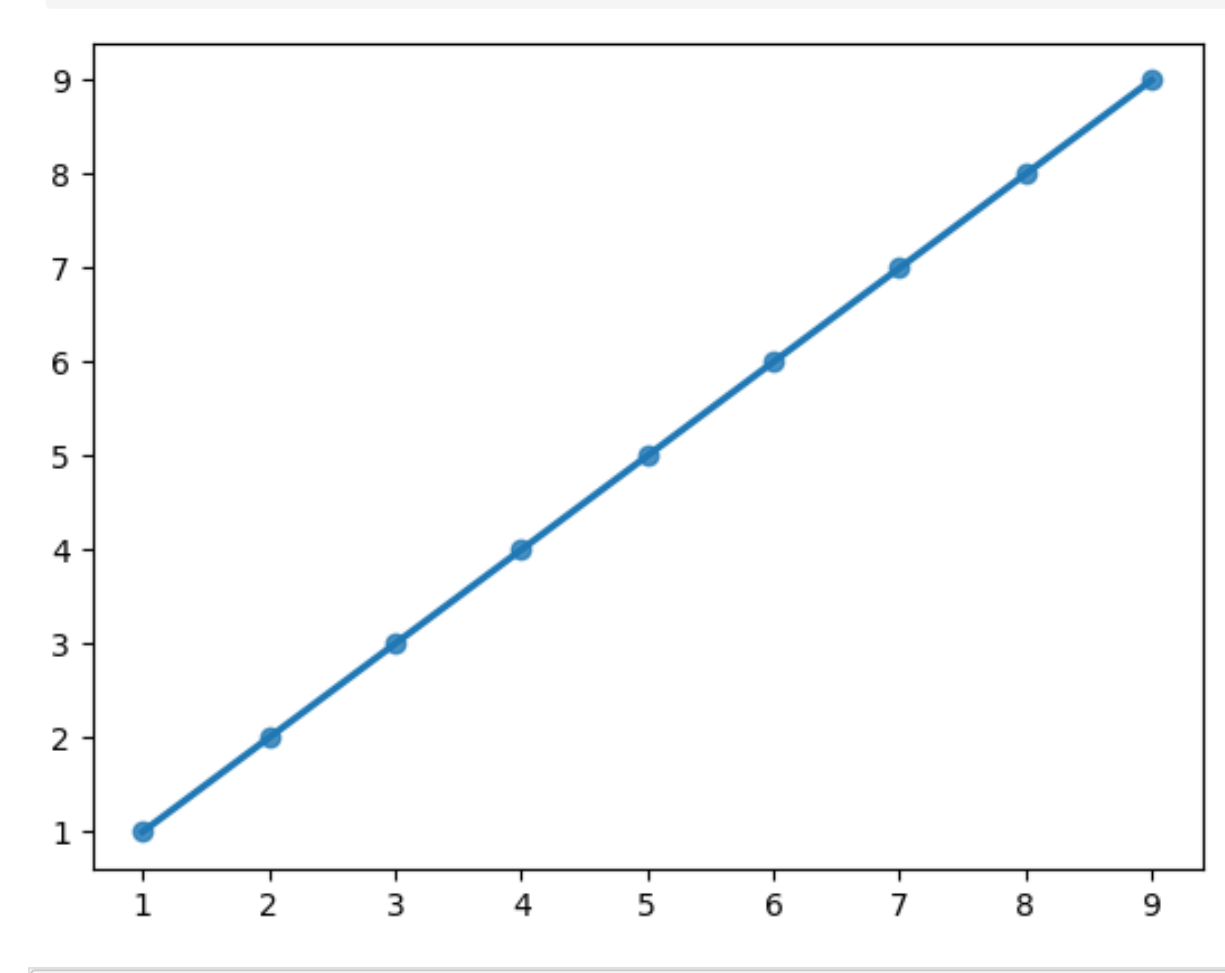
- draws a regression line of best fit - same as `np.polyfit()`

In [3…

```
sns.regplot(x = x, y = x)
```

Out[3…

```
<Axes: >
```

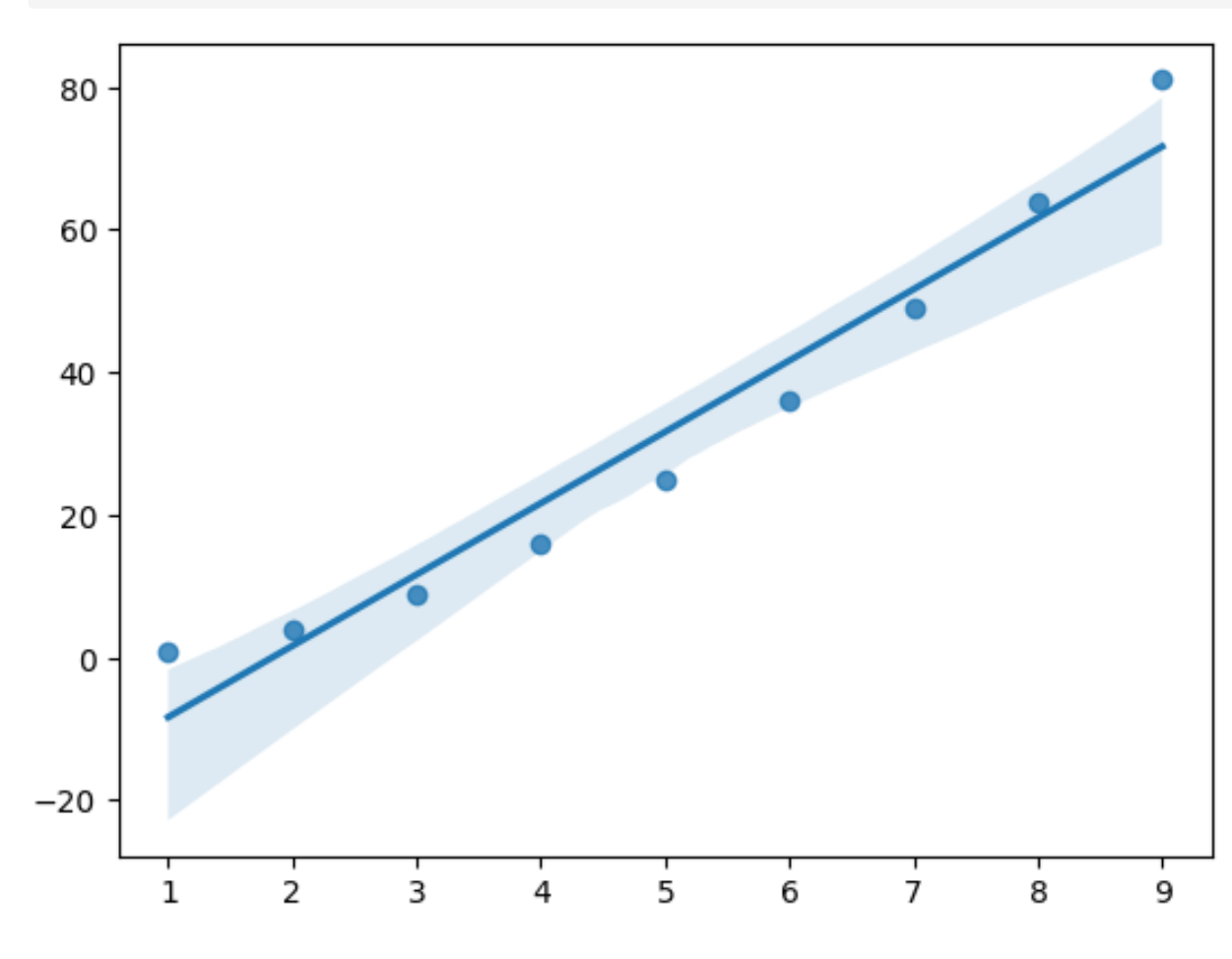


In […

```
sns.regplot(x = x, y = y)
```

Out[…

```
<Axes: >
```



`np.polyfit()`

- line of best fit - for drawing on plt plots

In […

```
a, b = np.polyfit(x, y, 1)
plt.plot(x, a*x+b)
```

What about quadratic and polynomial data?

In [3…

```
x1 = x
x2 = x
y1 = x**2
y2 = x**3
```

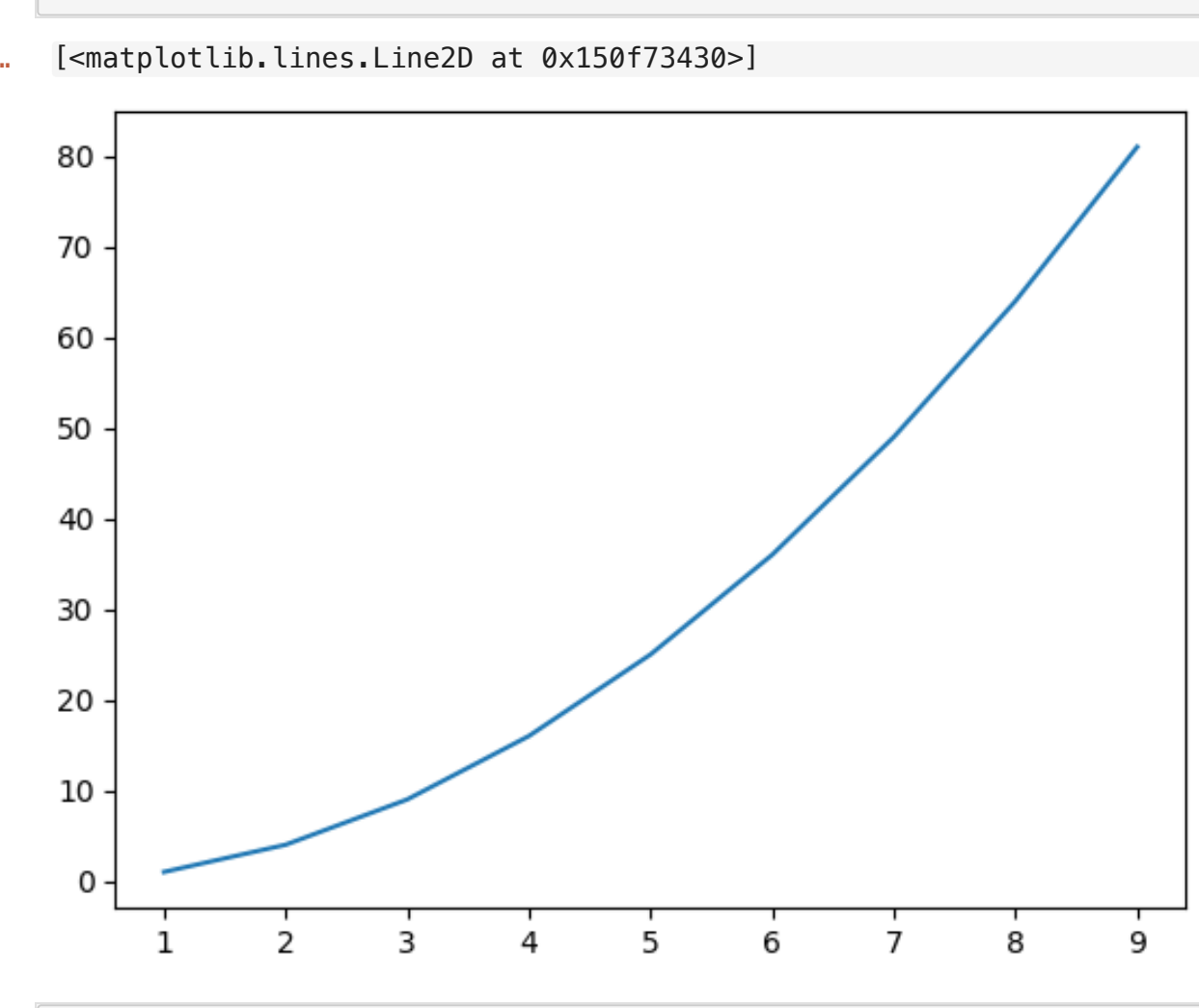
`plt.plot()` can draw us a polynomial regression curve

In [4…

```
plt.plot(x,y)
```

Out[4…

```
[<matplotlib.lines.Line2D at 0x150f73430>]
```

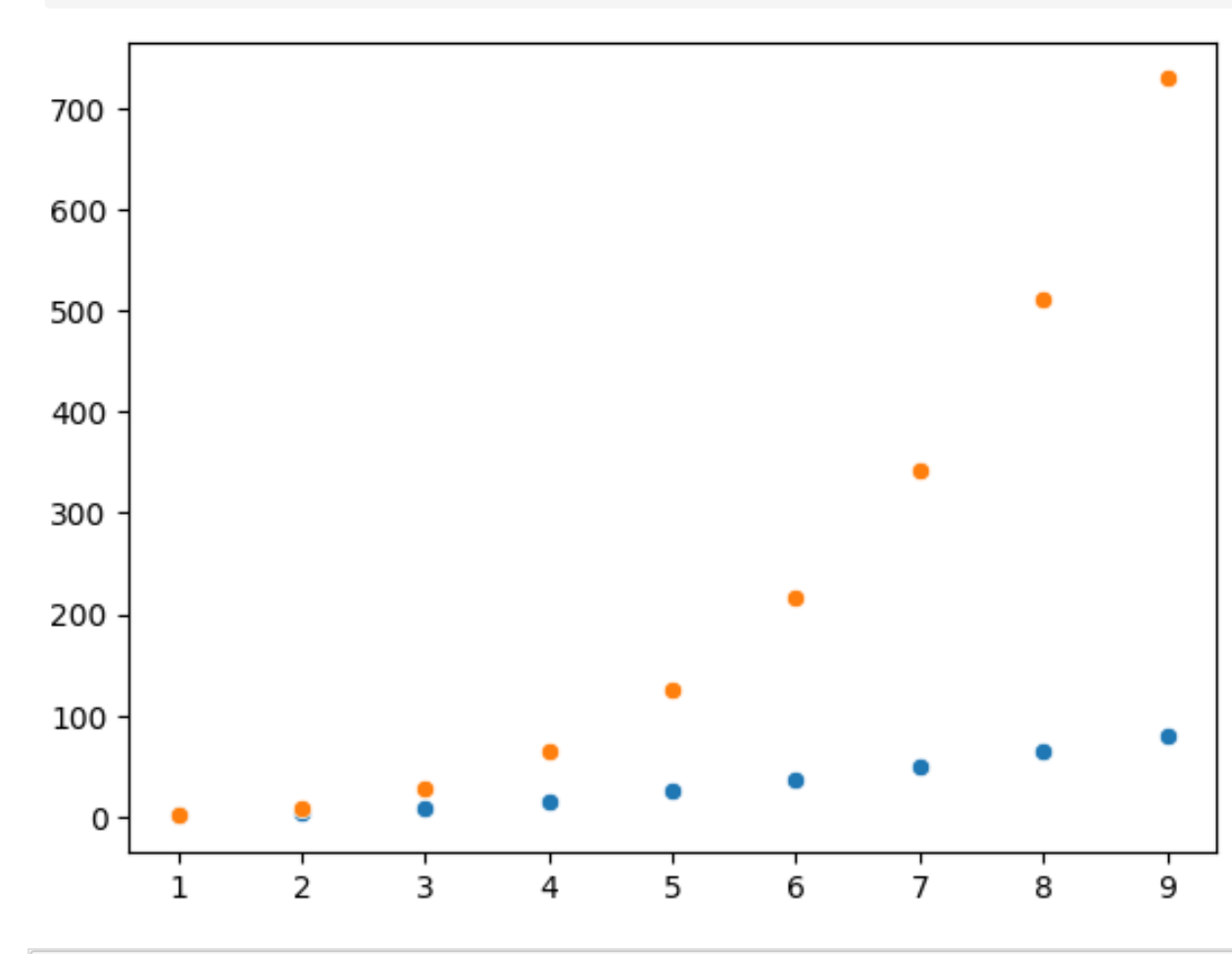


In [4…

```
sns.scatterplot(x=x1, y=y1)
sns.scatterplot(x=x2, y=y2)
```

Out[4…

```
<Axes: >
```

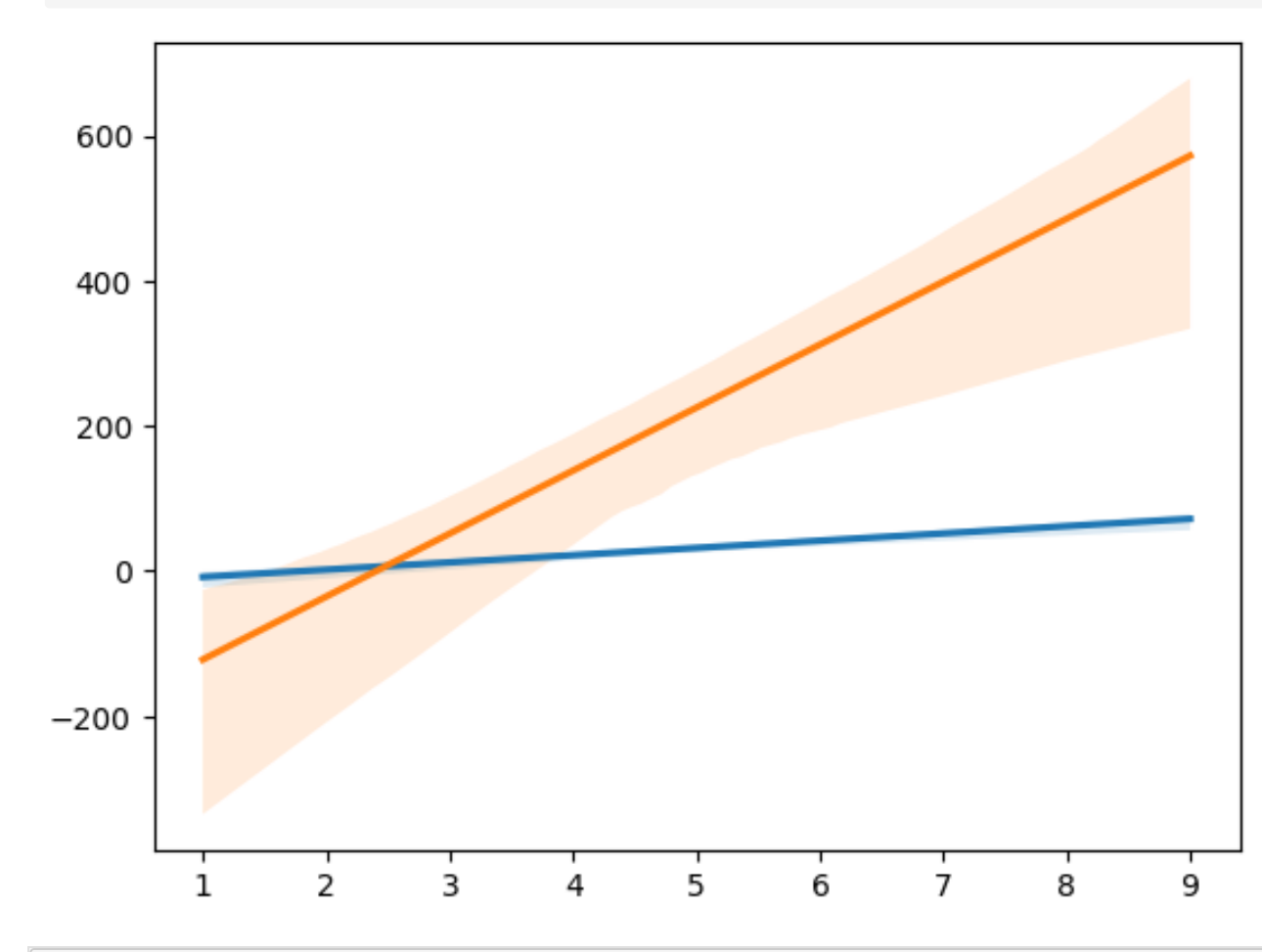


In [4…

```
sns.regplot(x=x1, y=y1, scatter=False)
sns.regplot(x=x2, y=y2, scatter=False)
```

Out[4…

```
<Axes: >
```



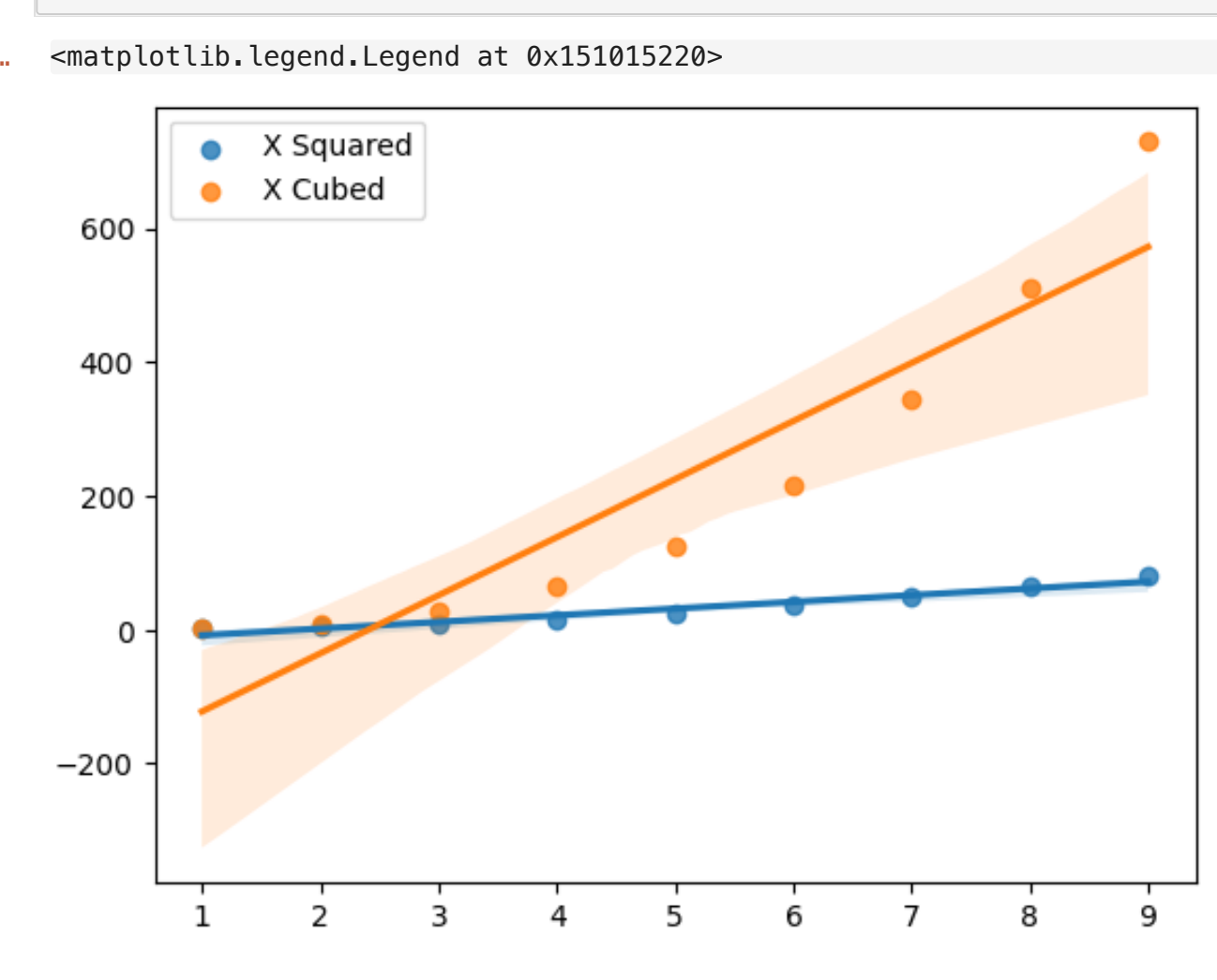
In [4…

```
sns.regplot(x=x1, y=y1, scatter=True, label="X Squared")
sns.regplot(x=x2, y=y2, scatter=True, label="X Cubed")

plt.legend()
```

Out[4…

```
<matplotlib.legend.Legend at 0x151015220>
```



`sns.regplot()` doesn't naturally handle polynomial regression

In [4…

```
df = pd.DataFrame({
    'x': [1, 2, 3, 4, 5],
    'y': [1.2, 1.9, 3.2, 3.8, 5.1]
})

df['x2'] = df['x'] ** 2

# Use regplot with x as the polynomial-transformed variable
sns.regplot(x='x', y='y', data=df, order=2)  # This works only for regplot via keyword `order`

plt.show()
```

```
/Users/nick/Library/Python/3.9/lib/python/site-packages/seaborn/regression.py:261: RankWarning: Polyfit may be poorly conditioned
  return np.polyval(np.polyfit(_x, _y, order), grid)
/Users/nick/Library/Python/3.9/lib/python/site-packages/seaborn/regression.py:261: RankWarning: Polyfit may be poorly conditioned
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/Users/nick/Library/Python/3.9/lib/python/site-packages/seaborn/regression.py:261: RankWarning: Polyfit may be poorly conditioned
  return np.polyval(np.polyfit(_x, _y, order), grid)
/Users/nick/Library/Python/3.9/lib/python/site-packages/seaborn/regression.py:261: RankWarning: Polyfit may be poorly conditioned
  return np.polyval(np.polyfit(_x, _y, order), grid)
/Users/nick/Library/Python/3.9/lib/python/site-packages/seaborn/regression.py:261: RankWarning: Polyfit may be poorly conditioned
  return np.polyval(np.polyfit(_x, _y, order), grid)
/Users/nick/Library/Python/3.9/lib/python/site-packages/seaborn/regression.py:261: RankWarning: Polyfit may be poorly conditioned
  return np.polyval(np.polyfit(_x, _y, order), grid)
/Users/nick/Library/Python/3.9/lib/python/site-packages/seaborn/regression.py:261: RankWarning: Polyfit may be poorly conditioned
  return np.polyval(np.polyfit(_x, _y, order), grid)
/Users/nick/Library/Python/3.9/lib/python/site-packages/seaborn/regression.py:261: RankWarning: Polyfit may be poorly conditioned
  return np.polyval(np.polyfit(_x, _y, order), grid)
/Users/nick/Library/Python/3.9/lib/python/site-packages/seaborn/regression.py:261: RankWarning: Polyfit may be poorly conditioned
  return np.polyval(np.polyfit(_x, _y, order), grid)
/Users/nick/Library/Python/3.9/lib/python/site-packages/seaborn/regression.py:261: RankWarning: Polyfit may be poorly conditioned
  return np.polyval(np.polyfit(_x, _y, order), grid)
/Users/nick/Library/Python/3.9/lib/python/site-packages/seaborn/regression.py:261: RankWarning: Polyfit may be poorly conditioned
  return np.polyval(np.polyfit(_x, _y, order), grid)
/Users/nick/Library/Python/3.9/lib/python/site-packages/seaborn/regression.py:261: RankWarning: Polyfit may be poorly conditioned
  return np.polyval(np.polyfit(_x, _y, order), grid)
/Users/nick/Library/Python/3.9/lib/python/site-packages/seaborn/regression.py:261: RankWarning: Polyfit may be poorly conditioned
  return np.polyval(np.polyfit(_x, _y, order), grid)
/Users/nick/Library/Python/3.9/lib/python/site-packages/seaborn/regression.py:261: RankWarning: Polyfit may be poorly conditioned
  return np.polyval(np.polyfit(_x, _y, order), grid)
/Users/nick/Library/Python/3.9/lib/python/site-packages/seaborn/regression.py:261: RankWarning: Polyfit may be poorly conditioned
  return np.polyval(np.polyfit(_x, _y, order), grid)
/Users/nick/Library/Python/3.9/lib/python/site-packages/seaborn/regression.py:261: RankWarning: Polyfit may be poorly conditioned
  return np.polyval(np.polyfit(_x, _y, order), grid)
/Users/nick/Library/Python/3.9/lib/python/site-packages/seaborn/regression.py:261: RankWarning: Polyfit may be poorly conditioned
  return np.polyval(np.polyfit(_x, _y, order), grid)
/Users/nick/Library/Python/3.9/lib/python/site-packages/seaborn/regression.py:261: RankWarning: Polyfit may be poorly conditioned
  return np.polyval(np.polyfit(_x, _y, order), grid)
/Users/nick/Library/Python/3.9/lib/python/site-packages/seaborn/regression.py:261: RankWarning: Polyfit may be poorly conditioned
  return np.polyval(np.polyfit(_x, _y, order), grid)
/Users/nick/Library/Python/3.9/lib/python/site-packages/seaborn/regression.py:261: RankWarning: Polyfit may be poorly conditioned
  return np.polyval(np.polyfit(_x, _y, order), grid)
/Users/nick/Library/Python/3.9/lib/python/site-packages/seaborn/regression.py:261: RankWarning: Polyfit may be poorly conditioned
  return np.polyval(np.polyfit(_x, _y, order), grid)
/Users/nick/Library/Python/3.9/lib/python/site-packages/seaborn/regression.py:261: RankWarning: Polyfit may be poorly conditioned
  return np.polyval(np.polyfit(_x, _y, order), grid)
/Users/nick/Library/Python/3.9/lib/python/site-packages/seaborn/regression.py:261: RankWarning: Polyfit may be poorly conditioned
  return np.polyval(np.polyfit(_x, _y, order), grid)
/Users/nick/Library/Python/3.9/lib/python/site-packages/seaborn/regression.py:261: RankWarning: Polyfit may be poorly conditioned
  return np.polyval(np.polyfit(_x, _y, order), grid)
/Users/nick/Library/Python/3.9/lib/python/site-packages/seaborn/regression.py:261: RankWarning: Polyfit may be poorly conditioned
  return np.polyval(np.polyfit(_x, _y, order), grid)
/Users/nick/Library/Python/3.9/lib/python/site-packages/seaborn/regression.py:261: RankWarning: Polyfit may be poorly conditioned
  return np.polyval(np.polyfit(_x, _y, order), grid)
/Users/nick/Library/Python/3.9/lib/python/site-packages/seaborn/regression.py:261: RankWarning: Polyfit may be poorly conditioned
  return np.polyval(np.polyfit(_x, _y, order), grid)
/Users/nick/Library/Python/3.9/lib/python/site-packages/seaborn/regression.py:261: RankWarning: Polyfit may be poorly conditioned
  return np.polyval(np.polyfit(_x, _y, order), grid)
/Users/nick/Library/Python/3.9/lib/python/site-packages/seaborn/regression.py:261: RankWarning: Polyfit may be poorly conditioned
  return np.polyval(np.polyfit(_x, _y, order), grid)
/Users/nick/Library/Python/3.9/lib/python/site-packages/seaborn/regression.py:261: RankWarning: Polyfit may be poorly conditioned
  return np.polyval(np.polyfit(_x, _y, order), grid)
/Users/nick/Library/Python/3.9/lib/python/site-packages/seaborn/regression.py:261: RankWarning: Polyfit may be poorly conditioned
  return np.polyval(np.polyfit(_x, _y, order), grid)
/Users/nick/Library/Python/3.9/lib/python/site-packages/seaborn/regression.py:261: RankWarning: Polyfit may be poorly conditioned
  return np.polyval(np.polyfit(_x, _y, order), grid)
/Users/nick/Library/Python/3.9/lib/python/site-packages/seaborn/regression.py:261: RankWarning: Polyfit may be poorly conditioned
```
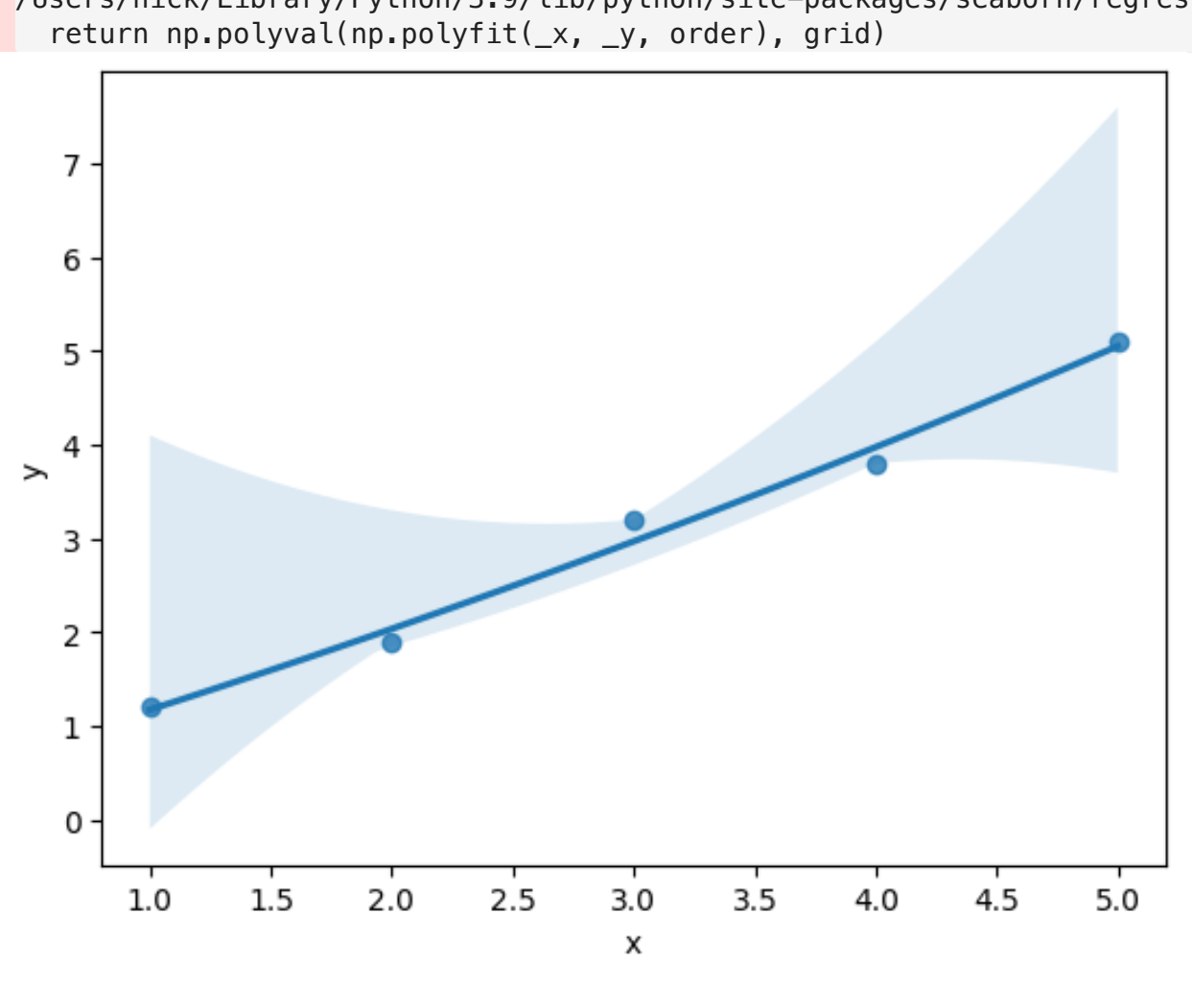
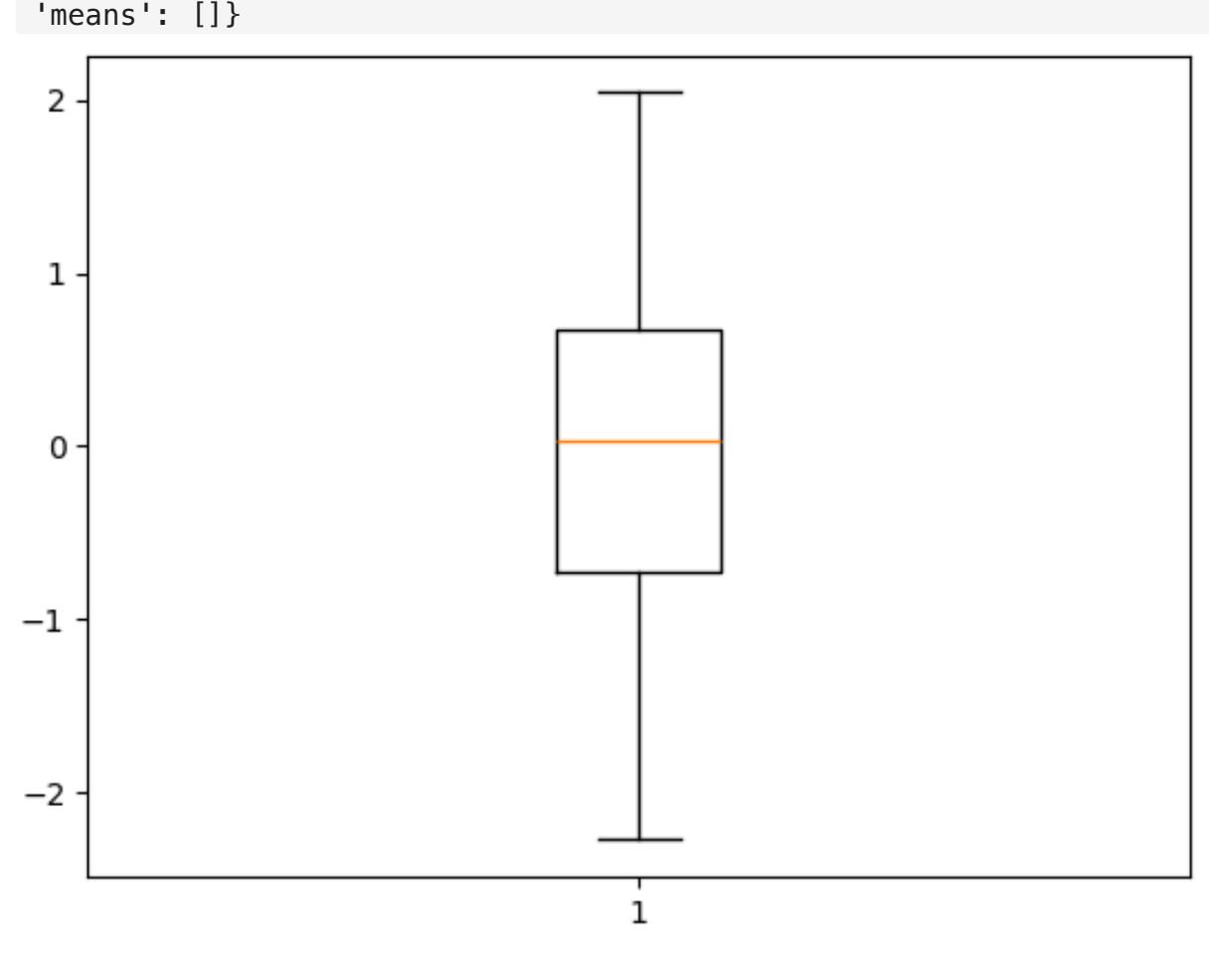

```
/Users/nick/Library/Python/3.9/lib/python/site-packages/seaborn/regression.py:201: RankWarning: Polyfit may be poorly conditioned
  return np.polyval(np.polyfit(_x, _y, order), grid)
```



`plt.boxplot()`

```
In [4…  std = 1
        data = [np.random.normal(0, std, 100)]
        plt.boxplot(data, vert = True)
```

```
Out[4…  {'whiskers': [<matplotlib.lines.Line2D at 0x15140ef70>,
          <matplotlib.lines.Line2D at 0x15141f250>],
         'caps': [<matplotlib.lines.Line2D at 0x15141f4f0>,
          <matplotlib.lines.Line2D at 0x15141f790>],
         'boxes': [<matplotlib.lines.Line2D at 0x15140ecd0>],
         'medians': [<matplotlib.lines.Line2D at 0x15141fa60>],
         'fliers': [<matplotlib.lines.Line2D at 0x15141fd00>],
         'means': []}
```



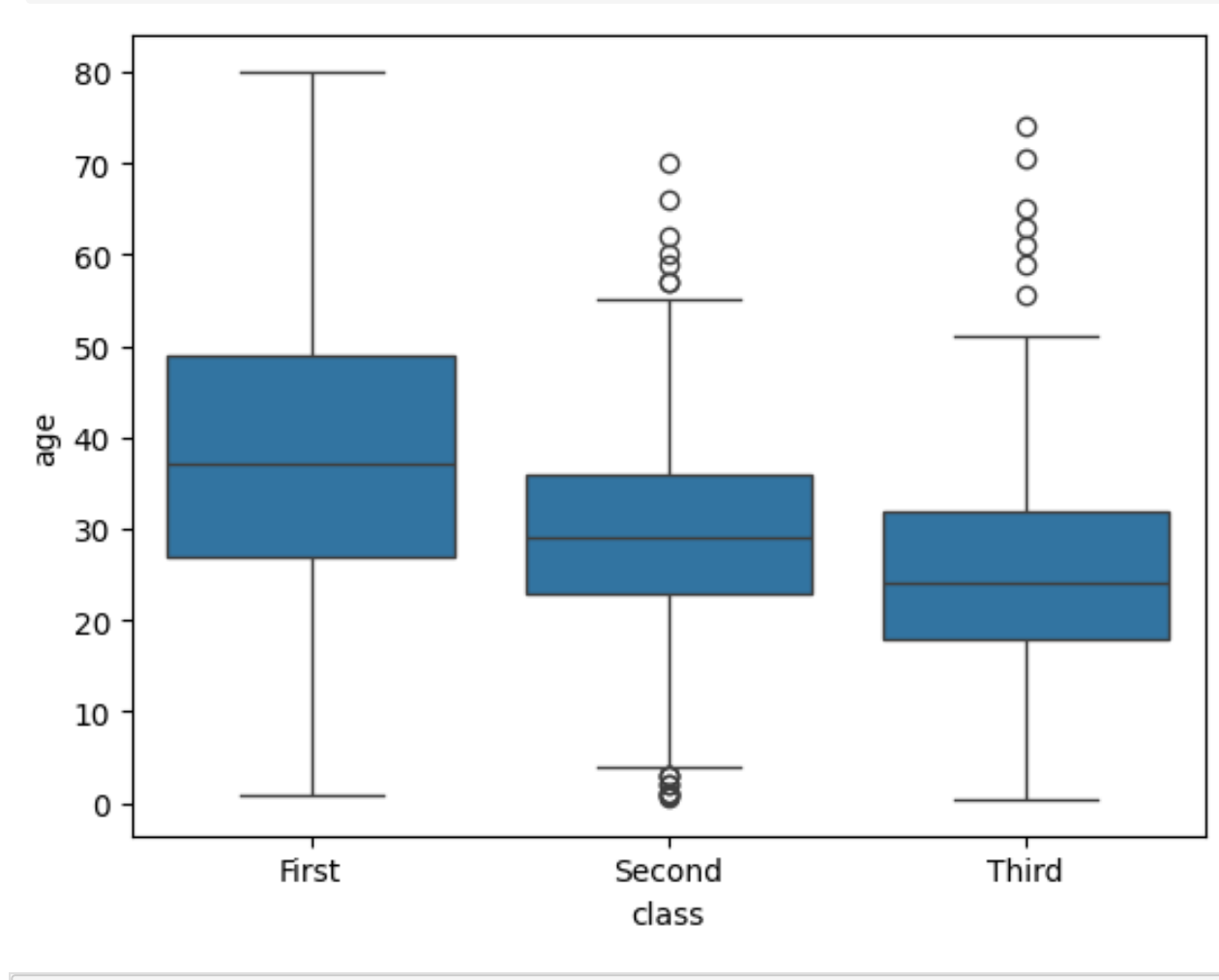
`sns.boxplot()`

```
In [4…  titanic.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 891 entries, 0 to 890
Data columns (total 15 columns):
 #   Column       Non-Null Count  Dtype
---  ------       --------------  -----
 0   survived     891 non-null    int64
 1   pclass       891 non-null    int64
 2   sex          891 non-null    object
 3   age          714 non-null    float64
 4   sibsp        891 non-null    int64
 5   parch        891 non-null    int64
 6   fare         891 non-null    float64
 7   embarked     889 non-null    object
 8   class        891 non-null    category
 9   who          891 non-null    object
 10  adult_male   891 non-null    bool
 11  deck         203 non-null    category
 12  embark_town  889 non-null    object
 13  alive        891 non-null    object
 14  alone        891 non-null    bool
dtypes: bool(2), category(2), float64(2), int64(4), object(5)
memory usage: 80.7+ KB
```

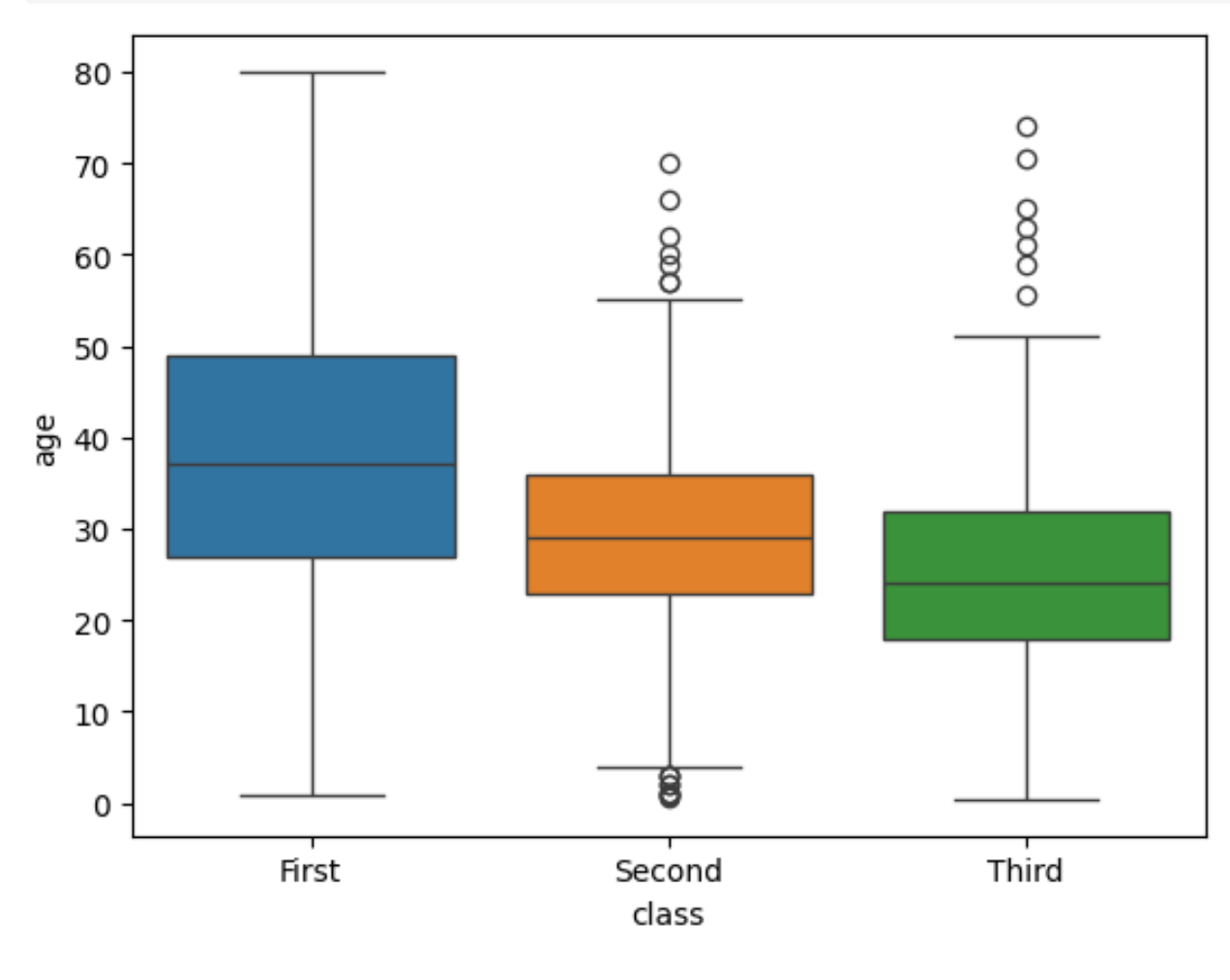
```
In [4…  sns.boxplot(x='class', y='age', data=titanic)
```

```
Out[4…  <Axes: xlabel='class', ylabel='age'>
```



```
In [5…  sns.boxplot(x='class', y='age', data=titanic, hue='class')
```

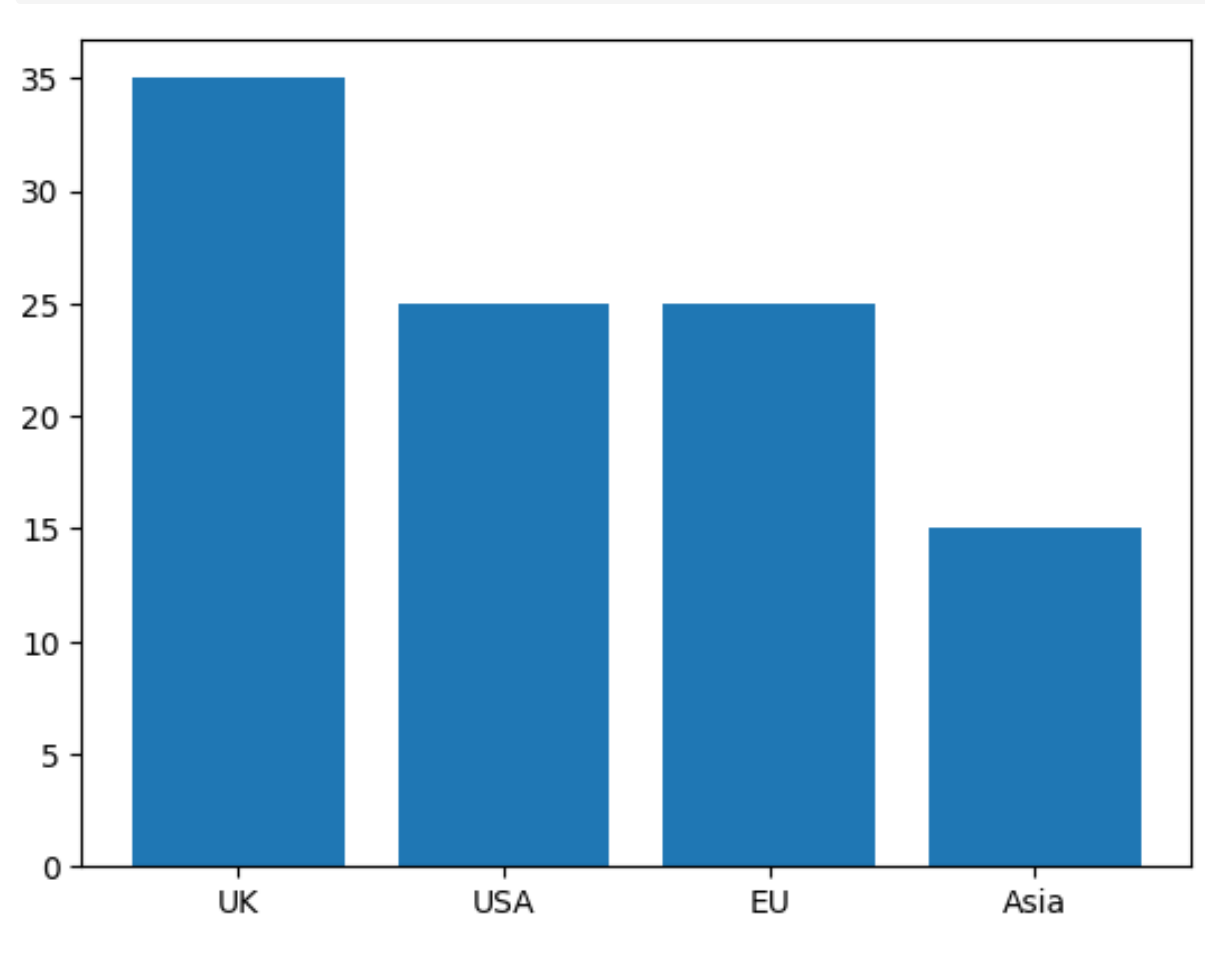
```
Out[5…  <Axes: xlabel='class', ylabel='age'>
```



`plt.bar()`

```
In [5…  data = [35, 25, 25, 15]
        labels = ["UK", "USA", "EU", "Asia"]

        plt.bar(labels, data)
```

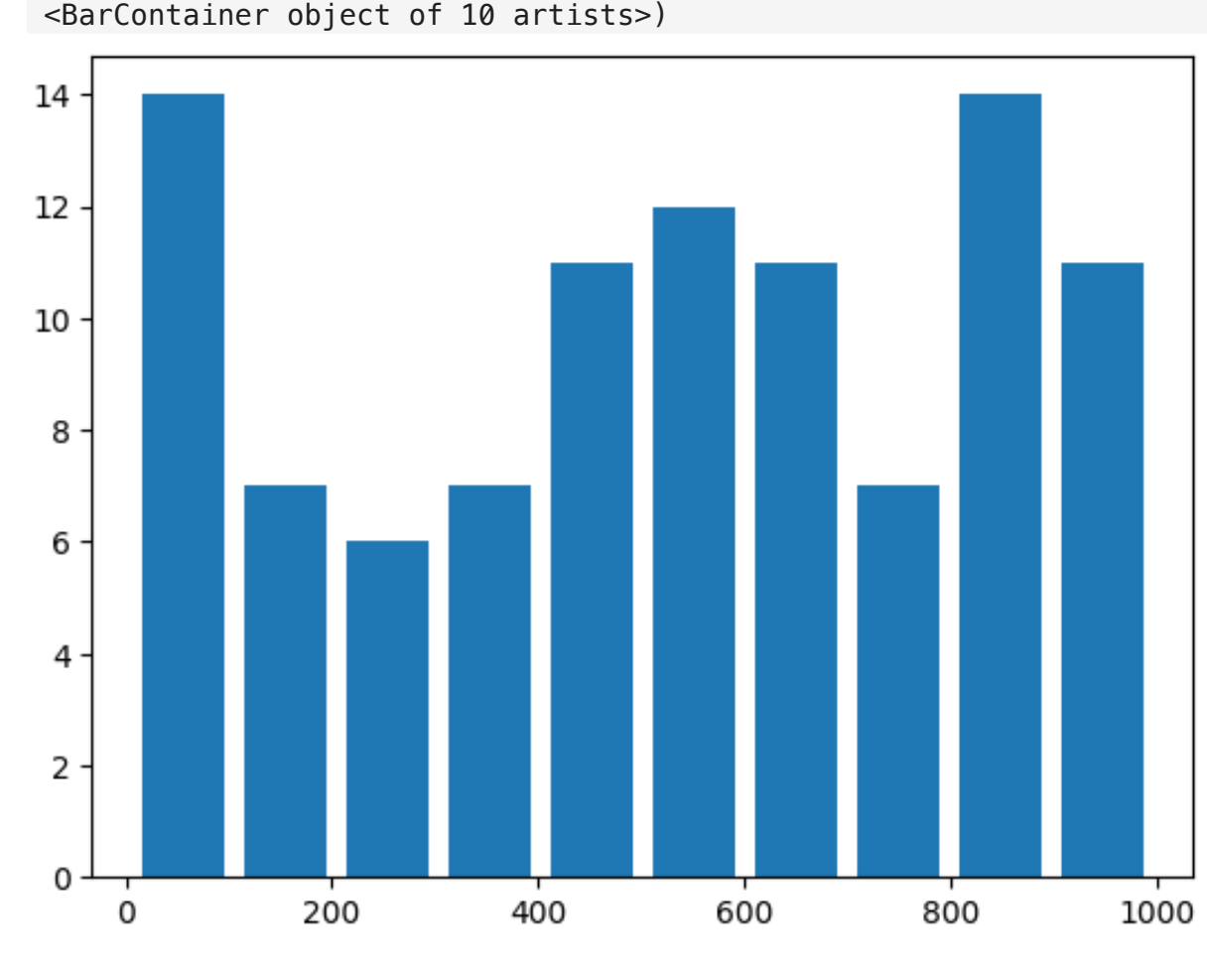
```
Out[5…  <BarContainer object of 4 artists>
```



`plt.hist()`

```
In [5…  from random import sample
        data = sample(range(1, 1000), 100)
        plt.hist(data, bins = 10, rwidth = 0.8)
```

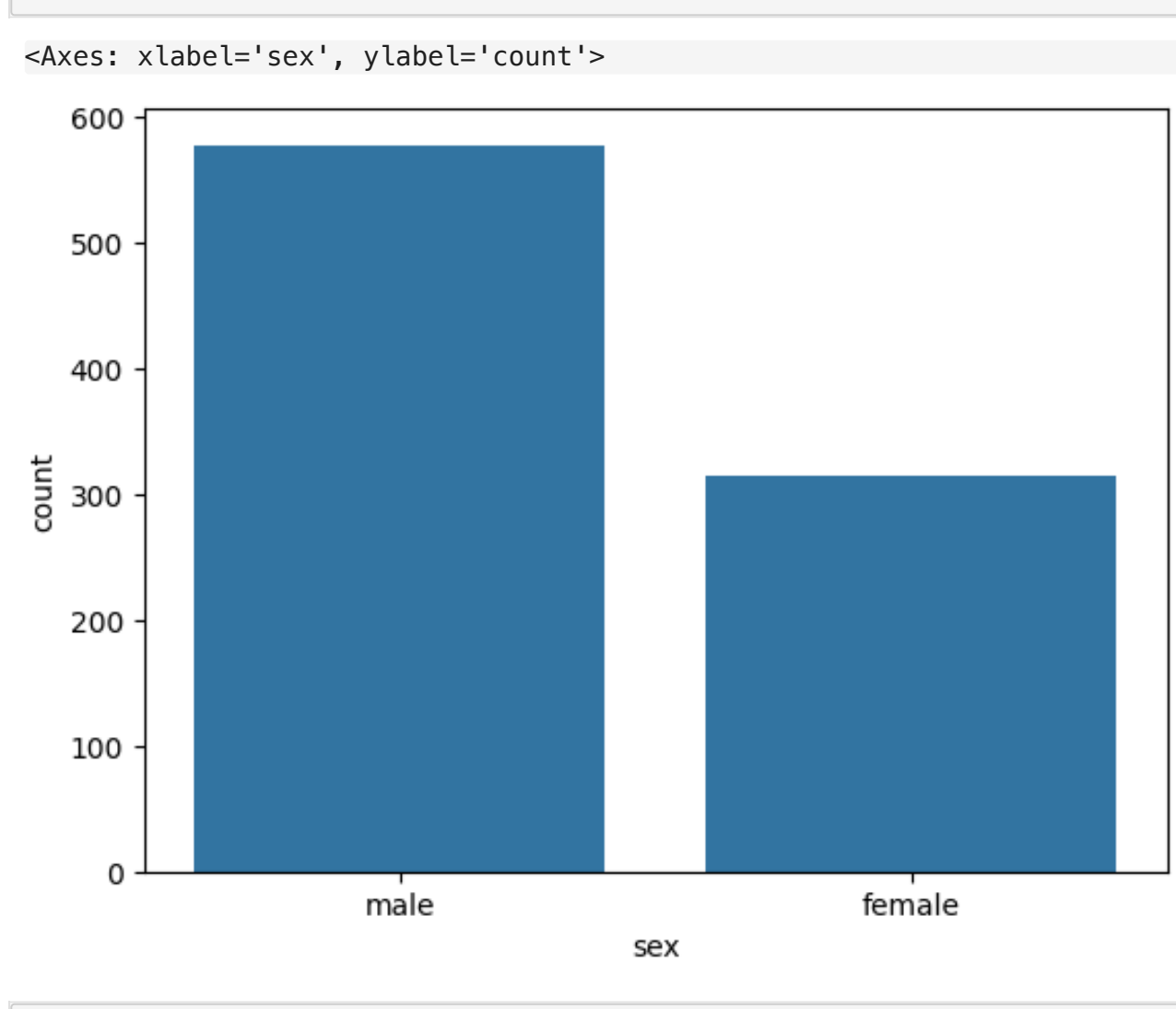
```
Out[5…  (array([14.,  7.,  6.,  7., 11., 12., 11.,  7., 14., 11.]),
         array([  6. , 105.1, 204.2, 303.3, 402.4, 501.5, 600.6, 699.7, 798.8,
                897.9, 997. ]),
         <BarContainer object of 10 artists>)
```



`sns.countplot()`

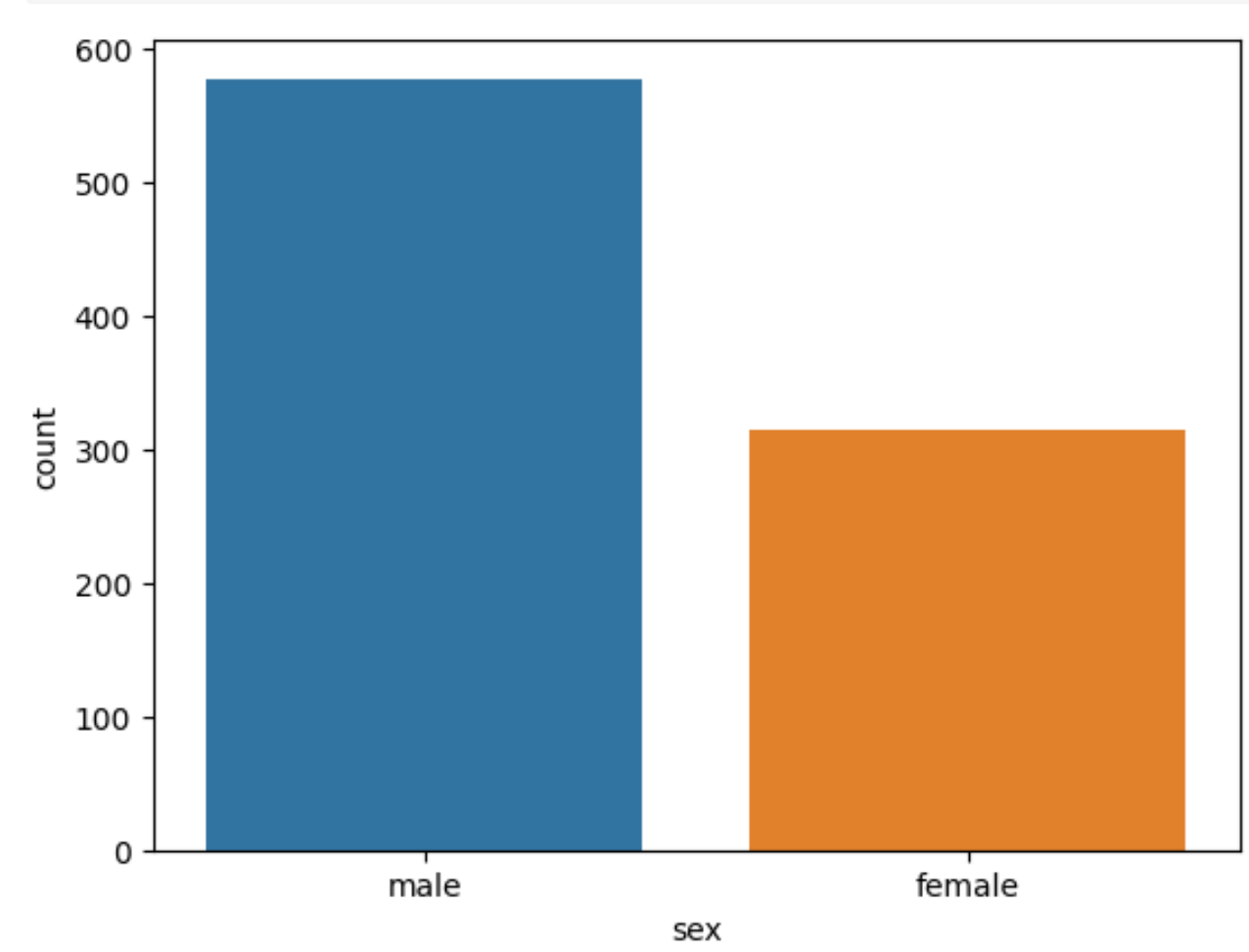
```
In [5…  sns.countplot(x='sex', data=titanic)
```

```
Out[5…  <Axes: xlabel='sex', ylabel='count'>
```



```
In [ …  sns.countplot(x='sex', data=titanic, hue='sex')
```

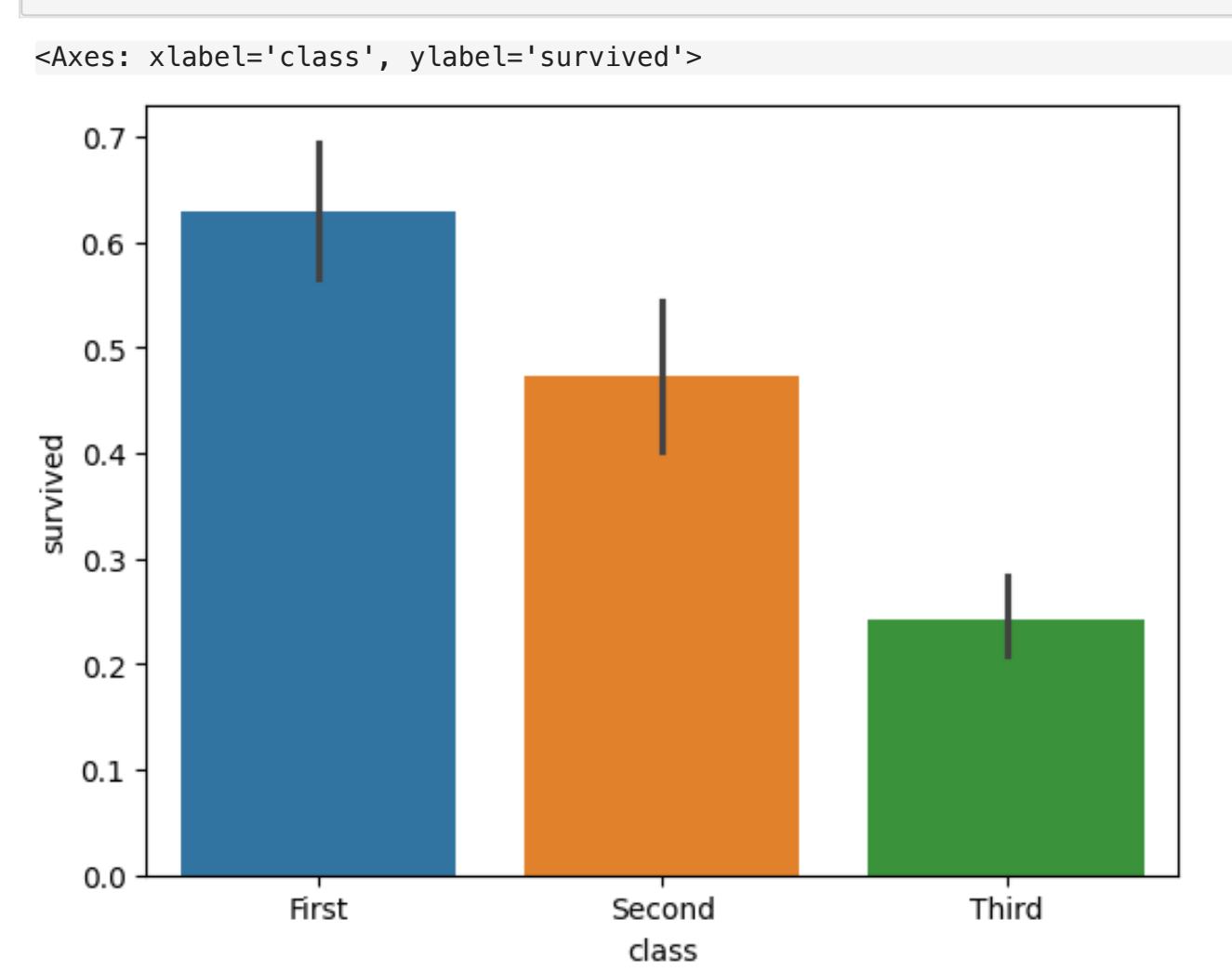
```
Out[ …  <Axes: xlabel='sex', ylabel='count'>
```



`sns.barplot()`

```
In [5…  sns.barplot(x='class', y='survived', data=titanic, hue="class")
```

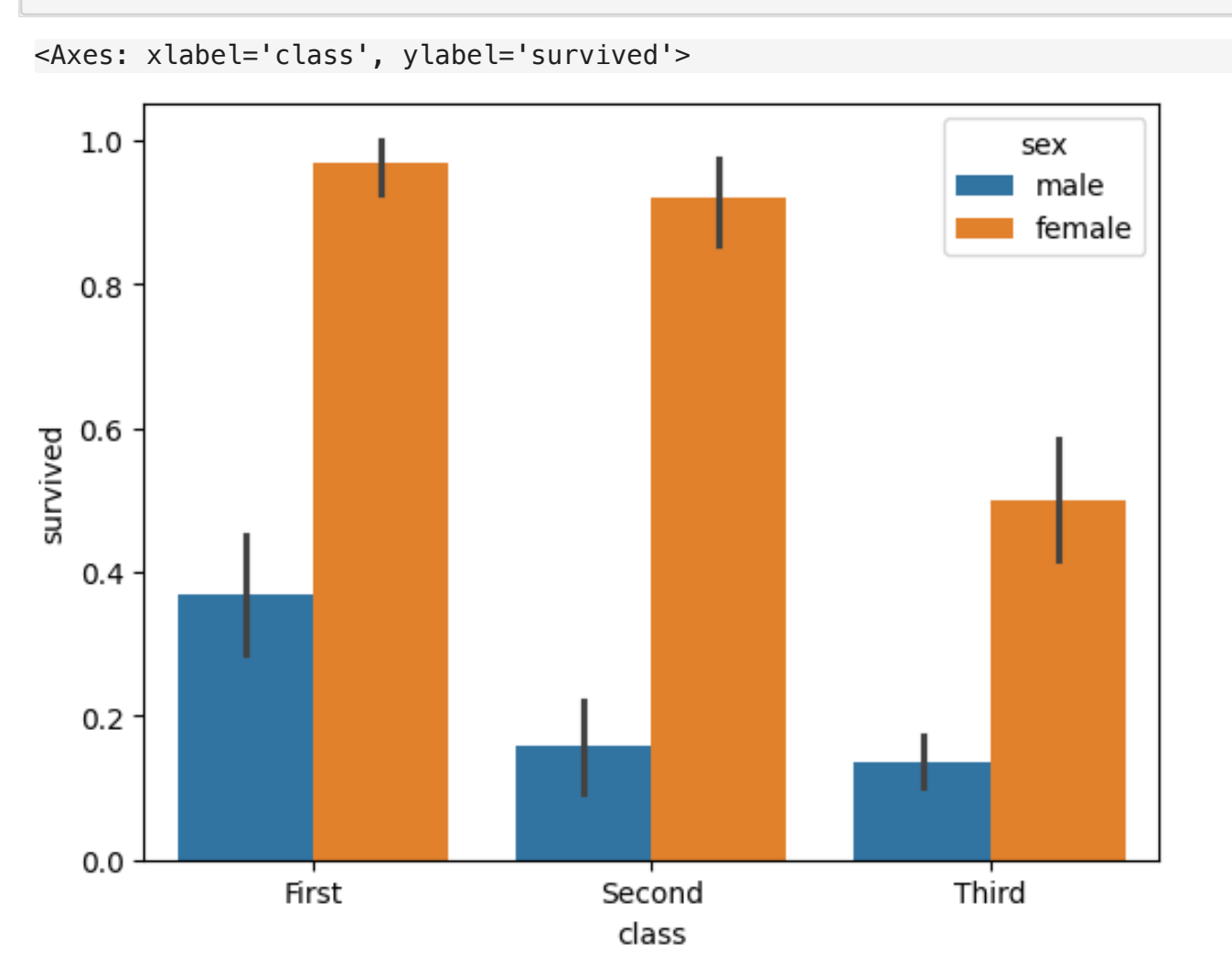
```
Out[5…  <Axes: xlabel='class', ylabel='survived'>
```



Same barplot, but can draw separate bars if we use a different column ('sex') for the `hue` attribute:

```
In [5…  sns.barplot(x='class', y ='survived', data=titanic, hue ='sex')
```

```
Out[5…  <Axes: xlabel='class', ylabel='survived'>
```

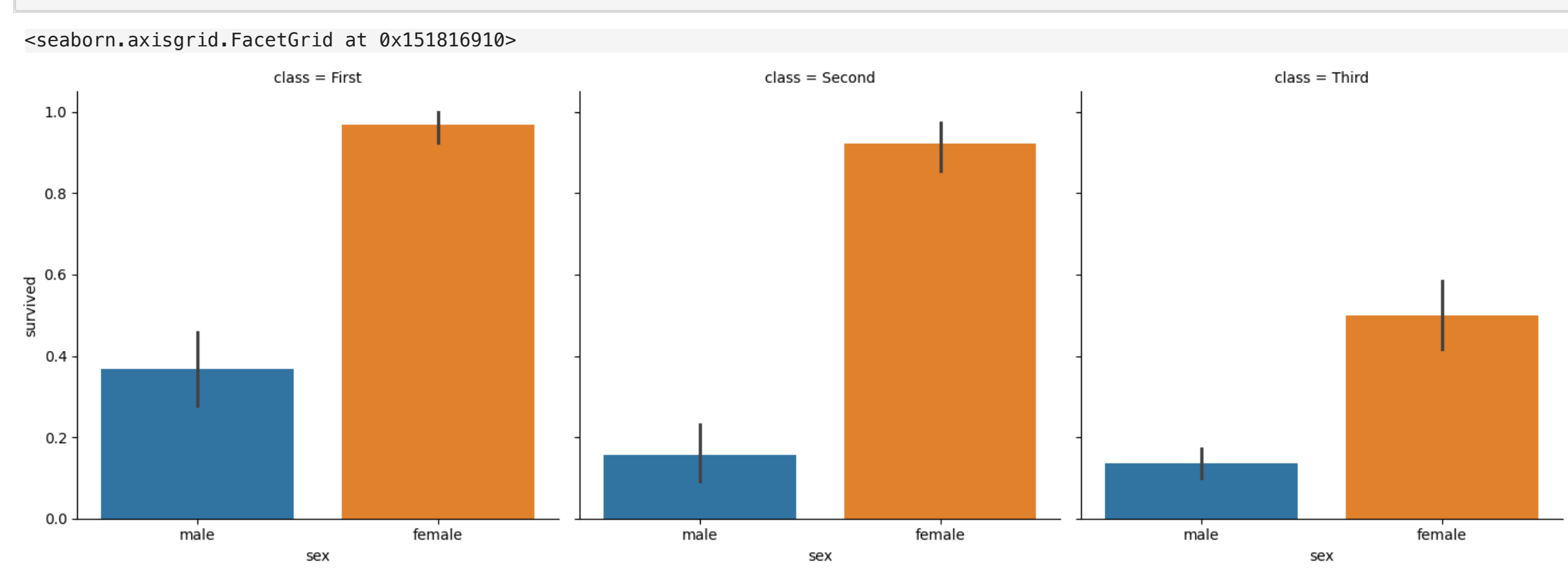


`sns.catplot()`

Just note that you need the `kind=bar` attribute below for `sns.catplot()`

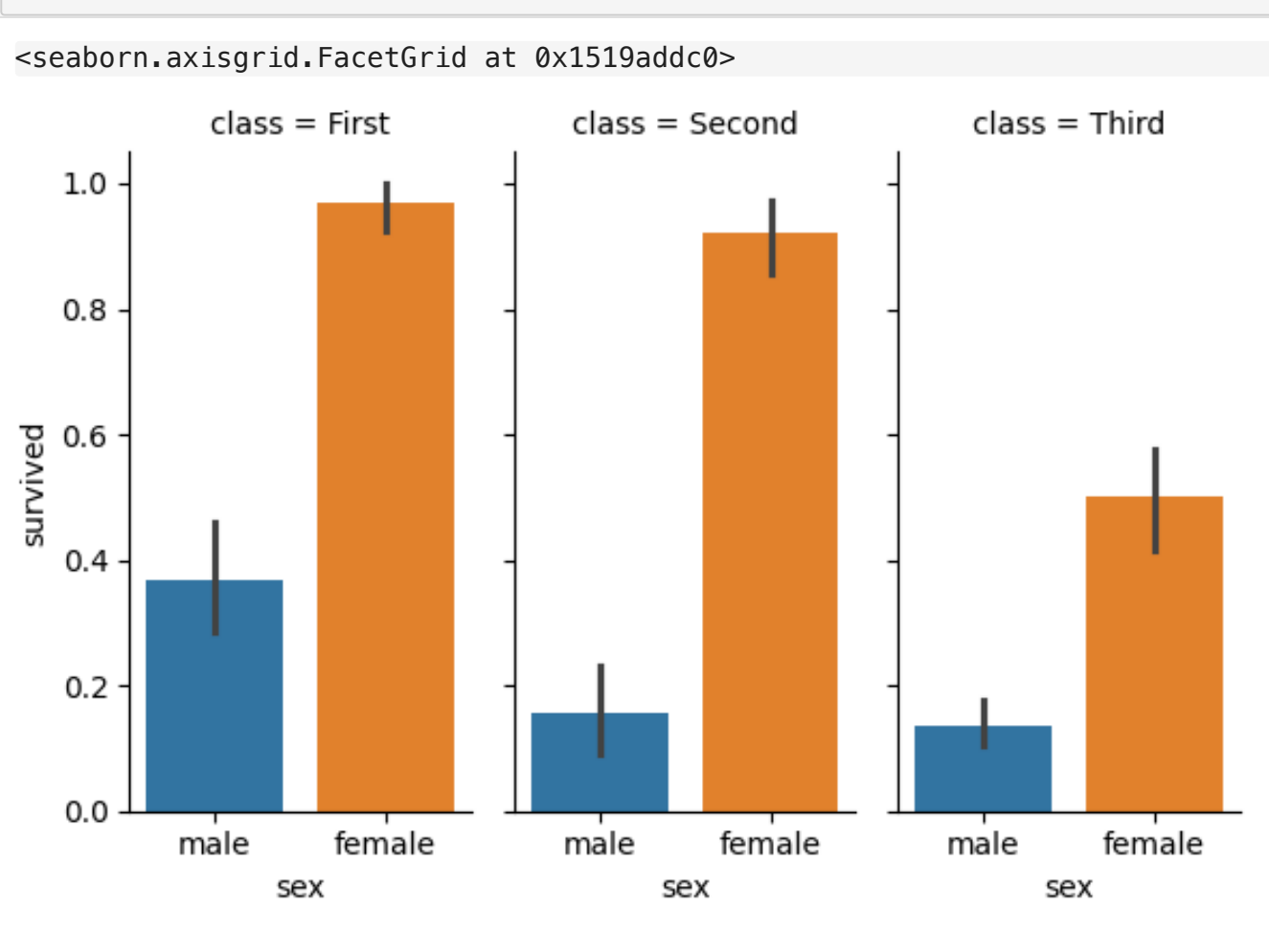
```
In [5…  sns.catplot(x='sex', y ='survived', data=titanic, col='class', hue ='sex', kind='bar')
```

```
Out[5…  <seaborn.axisgrid.FacetGrid at 0x151816910>
```



```
In [5…  sns.catplot(x='sex', y ='survived', data=titanic, col='class', hue ='sex', kind='bar', height=4, aspect=0.5)
```

```
Out[5…  <seaborn.axisgrid.FacetGrid at 0x1519addc0>
```

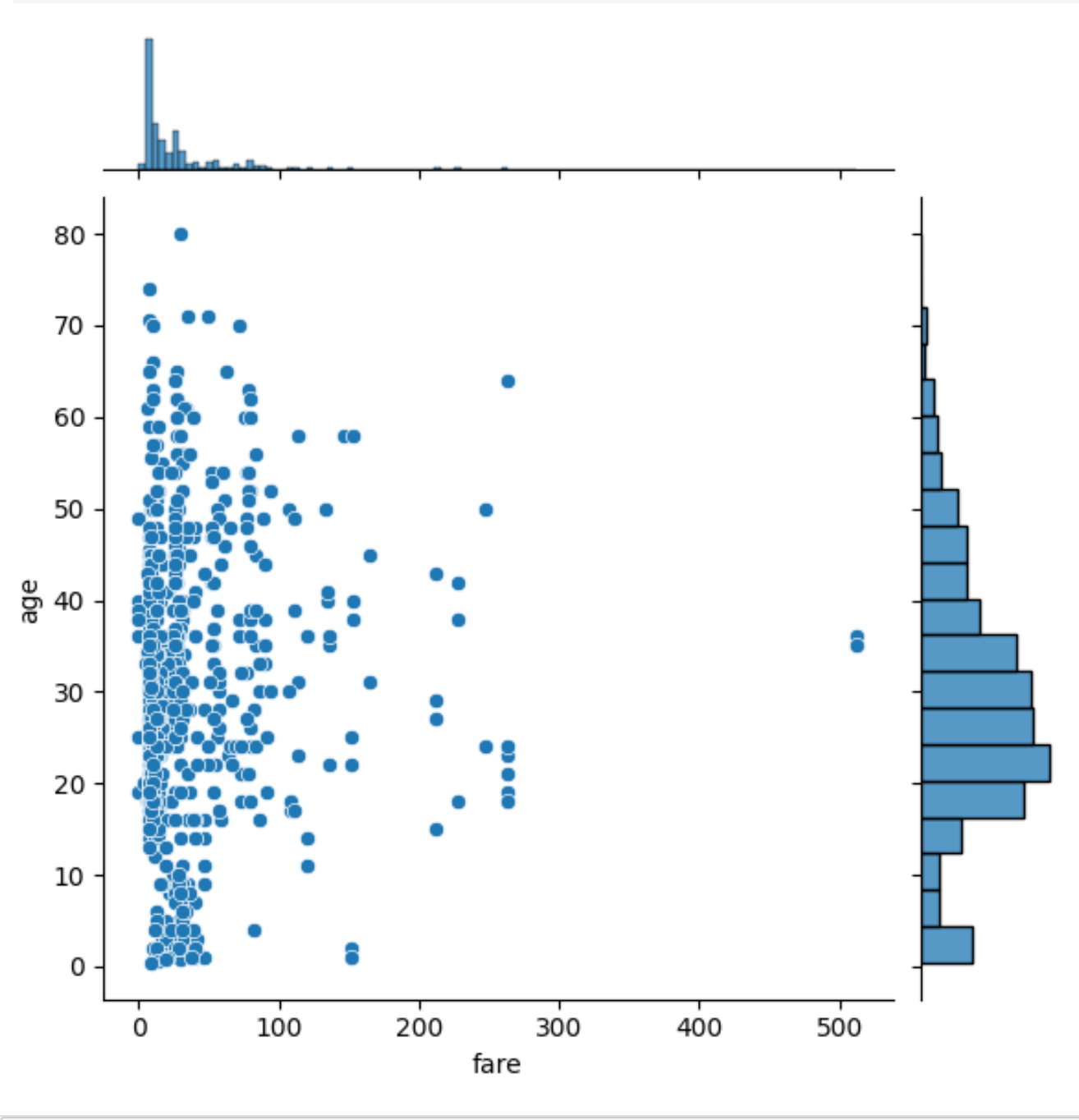


Seaborn's unique plotting functions:

- `sns.jointplot()` & `sns.distplot()` & `sns.pairplot()`
- `sns.swarmplot()`
- `sns.heatmap()`
- `sns.facetgrid()`

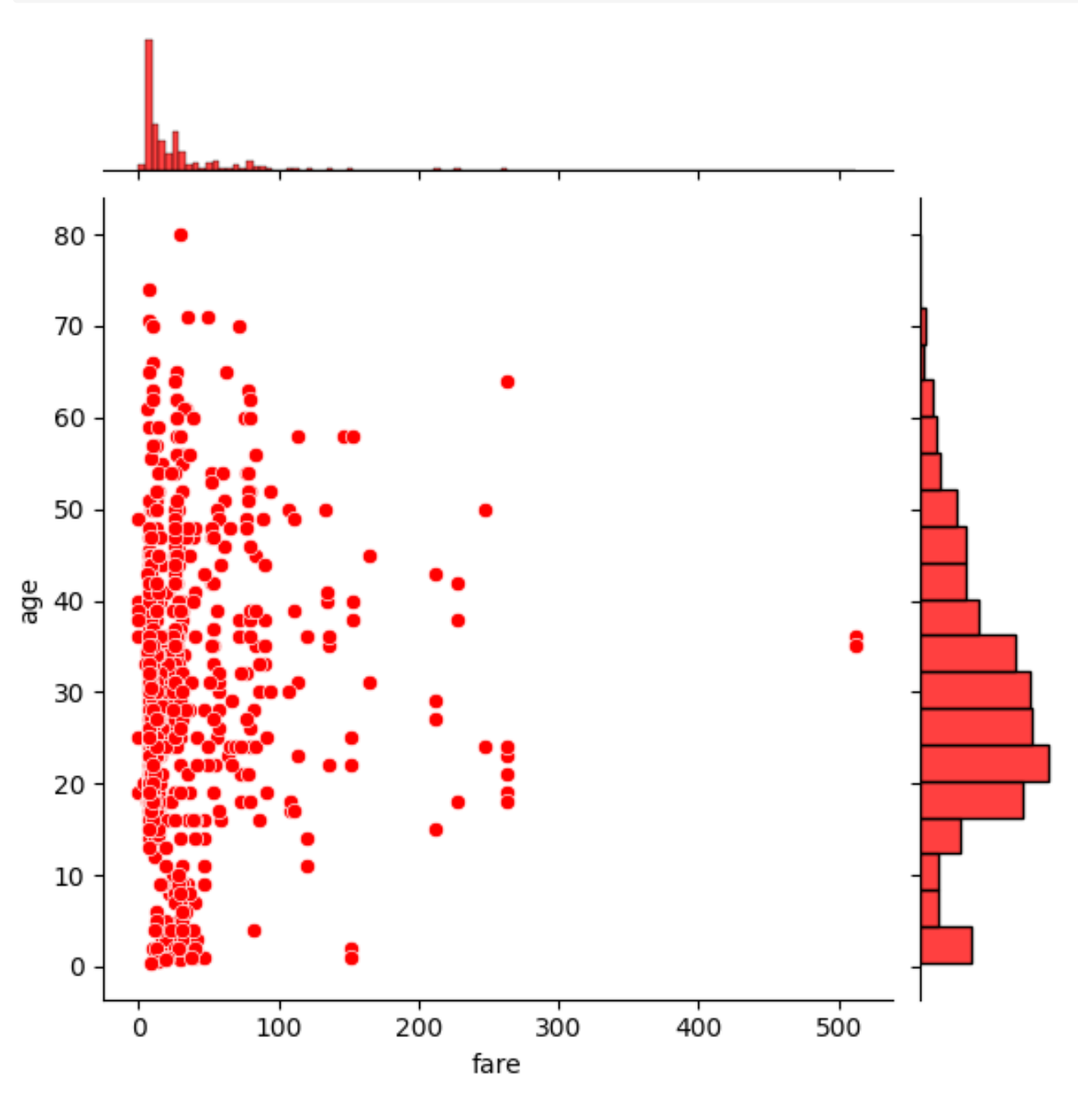
```
In [ …  sns.jointplot(x='fare', y='age', data=titanic)
```

```
Out[ …  <seaborn.axisgrid.JointGrid at 0x1519a0a60>
```



```
In [ …  sns.jointplot(x='fare', y='age', data=titanic, color='red')
```

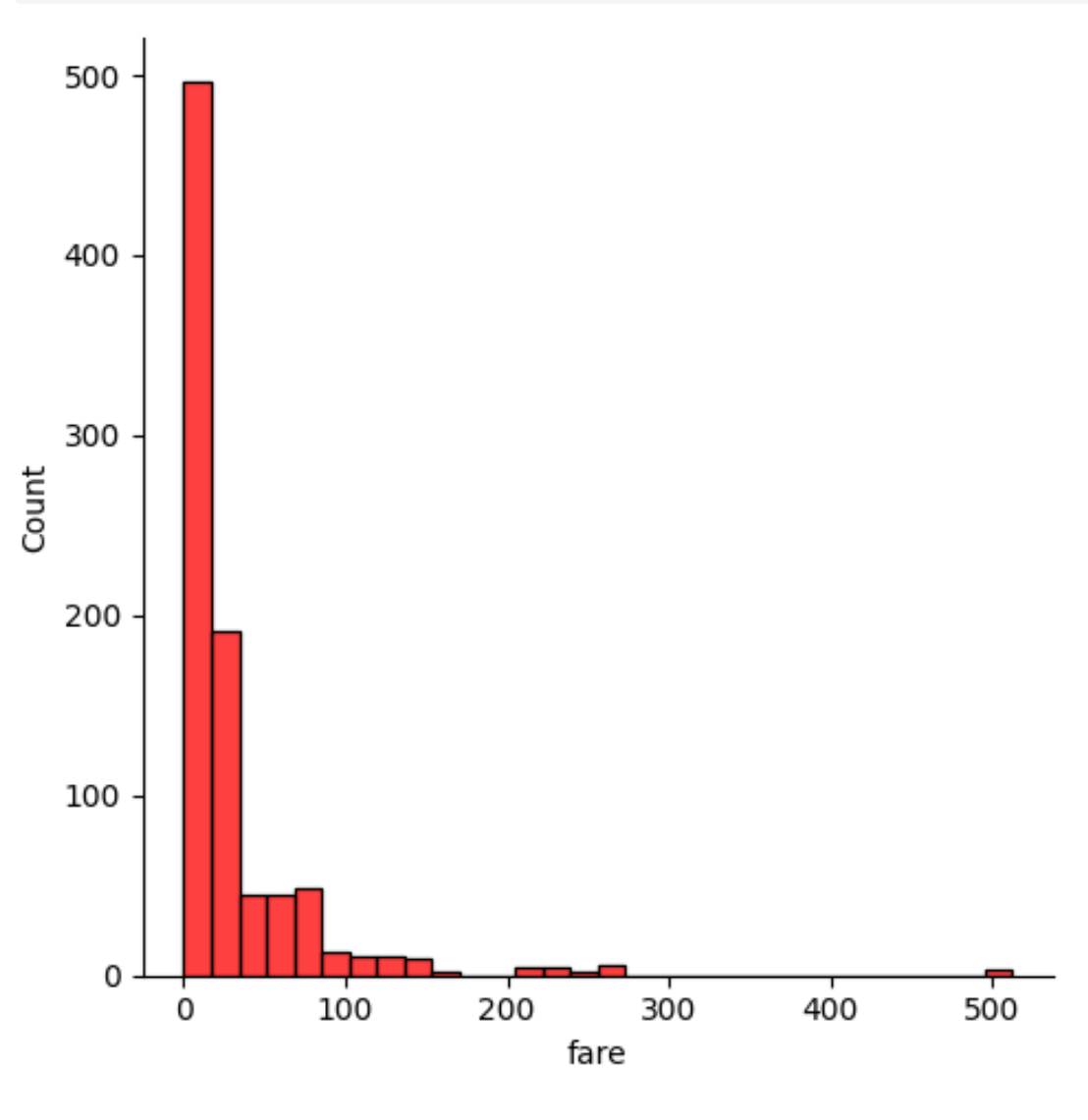
```
Out[ …  <seaborn.axisgrid.JointGrid at 0x150ce5af0>
```



`sns.distplot()`

```
In [6…  sns.displot(titanic['fare'], bins=30, kde=False, color='red')
```

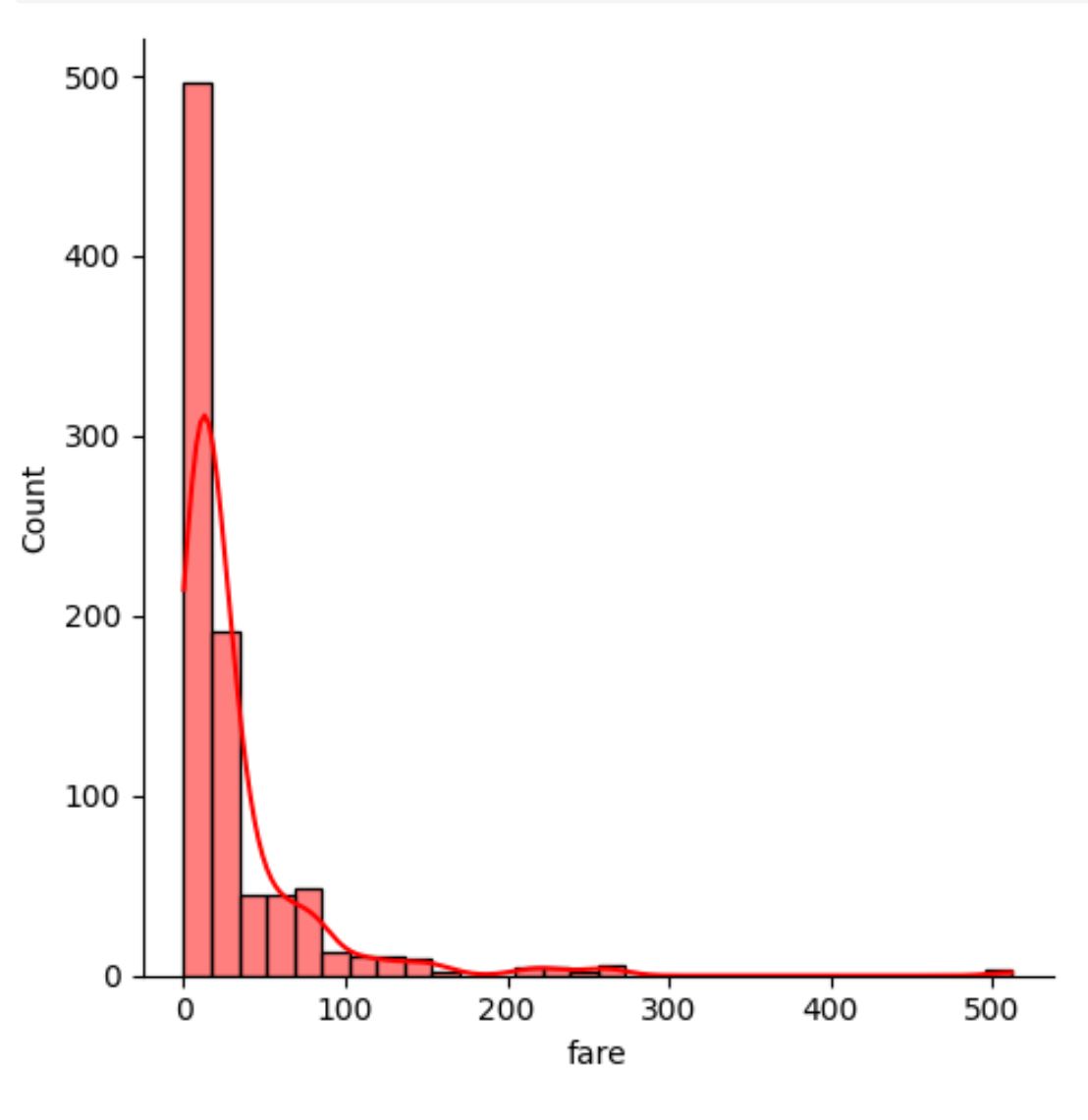
```
Out[6…  <seaborn.axisgrid.FacetGrid at 0x301dce3a0>
```



kernel density estimator (kde)

```
In [6…  sns.displot(titanic['fare'], bins=30, kde=True, color='red')
```

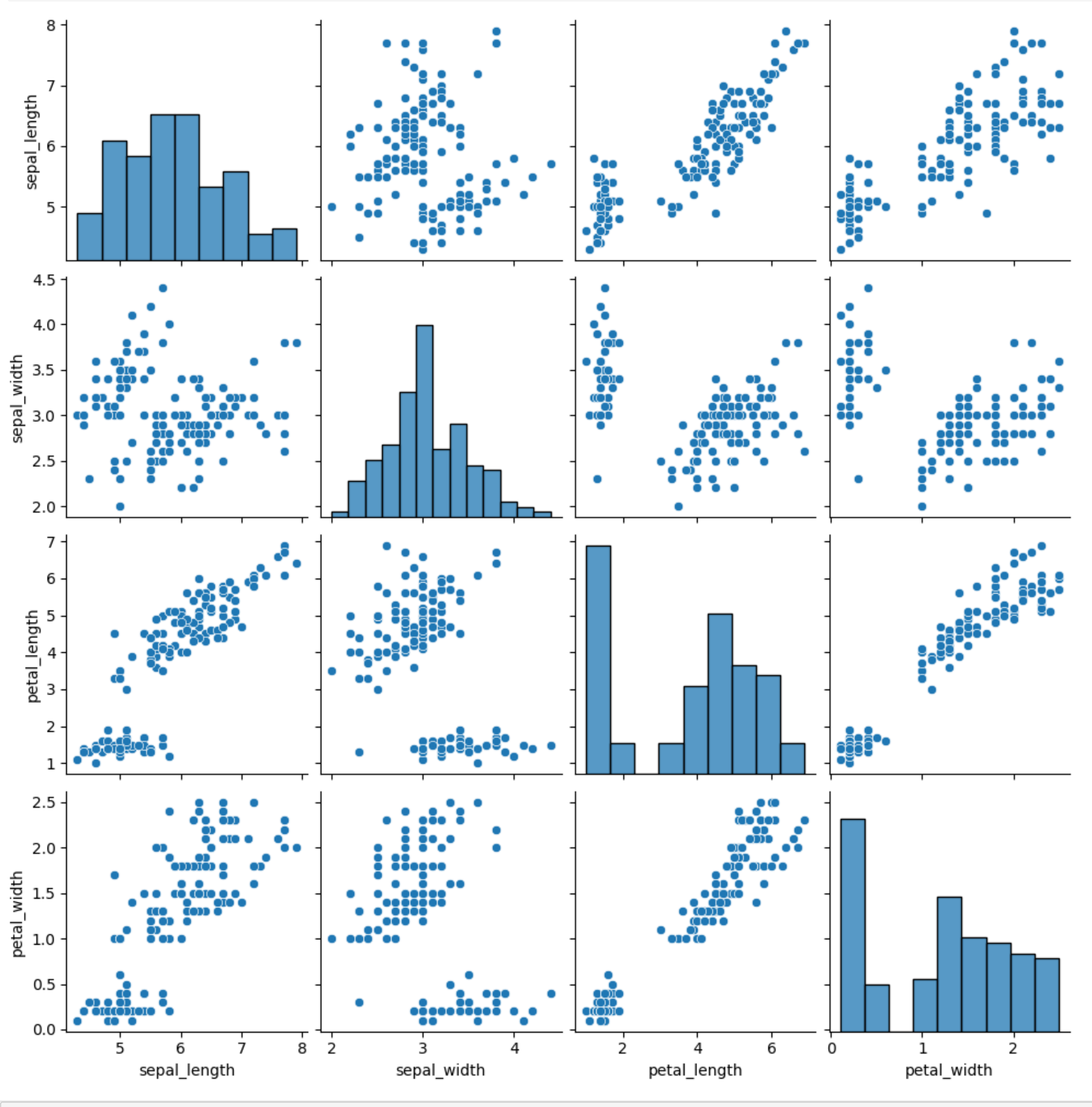
```
Out[6…  <seaborn.axisgrid.FacetGrid at 0x15282a640>
```



`sns.pairplot()`

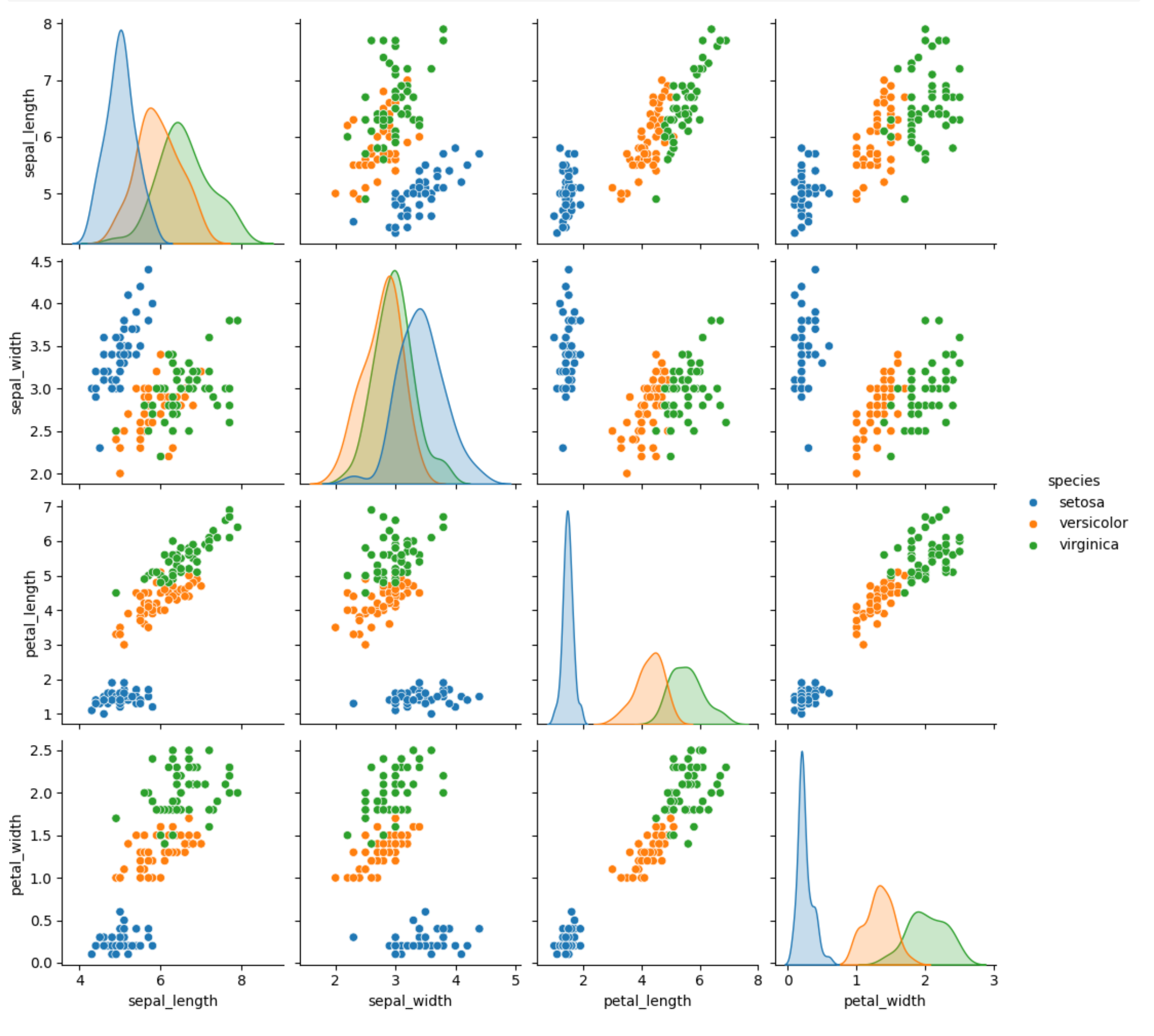
```
In [6…  sns.pairplot(iris, dropna=False)
```

```
Out[6…  <seaborn.axisgrid.PairGrid at 0x154917460>
```



```
In [6…  sns.pairplot(iris, dropna=False, hue='species')
```

```
Out[6…  <seaborn.axisgrid.PairGrid at 0x155113dc0>
```

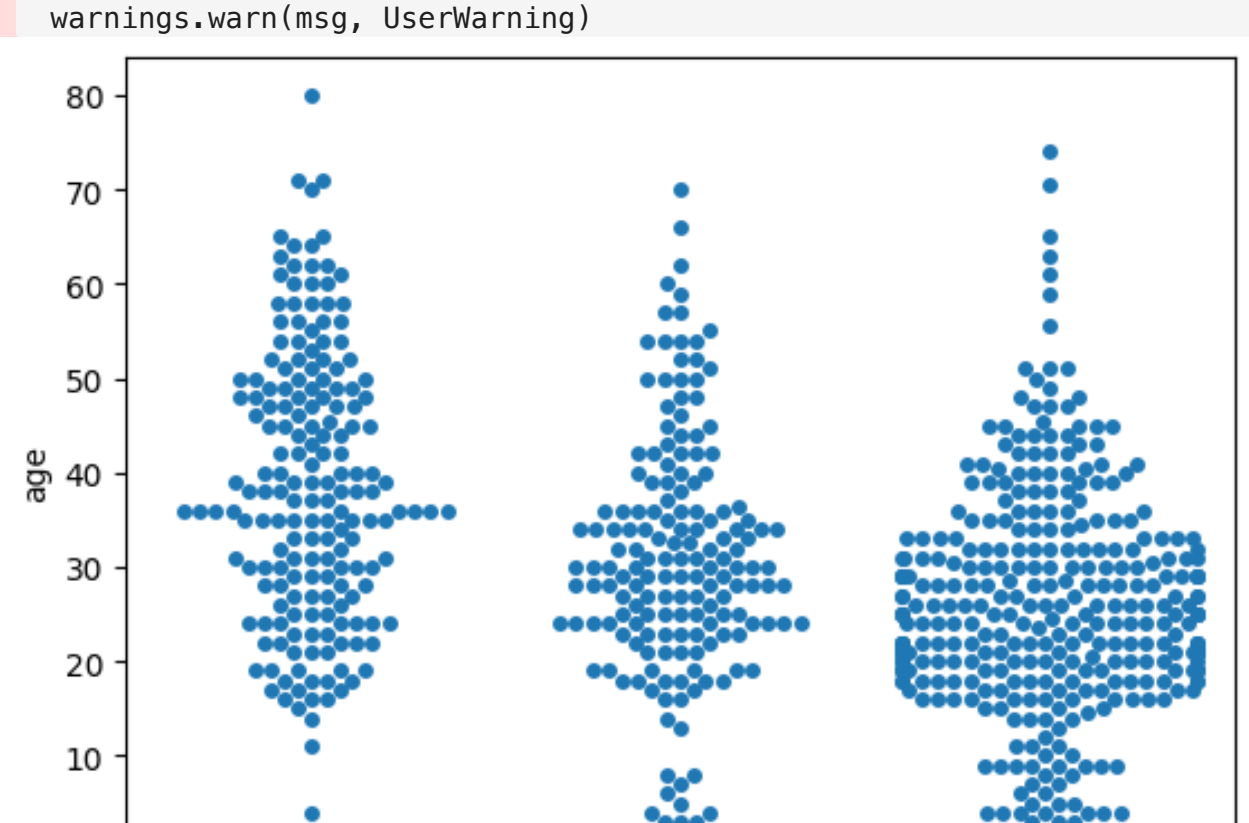


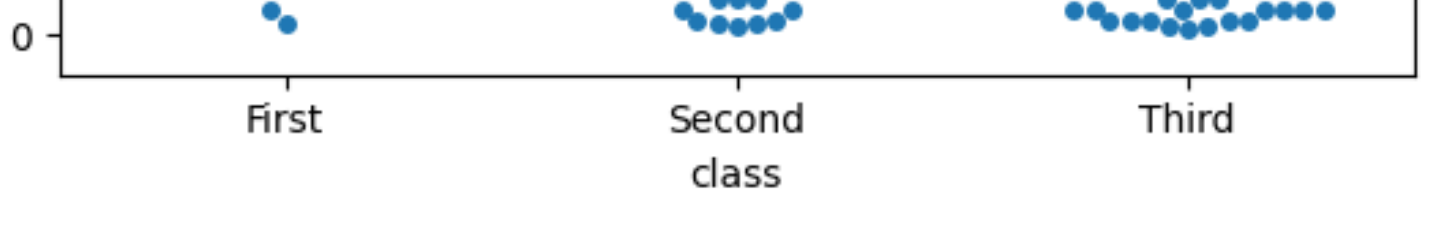
`sns.swarmplot()`

```
In [6…  sns.swarmplot(x='class', y='age', data=titanic)
```

```
Out[6…  <Axes: xlabel='class', ylabel='age'>
```

```
/Users/nick/Library/Python/3.9/lib/python/site-packages/seaborn/categorical.py:3399: UserWarning: 15.2% of the points cannot be placed; you may want to decrease the size of the markers or use stripplot.
  warnings.warn(msg, UserWarning)
```

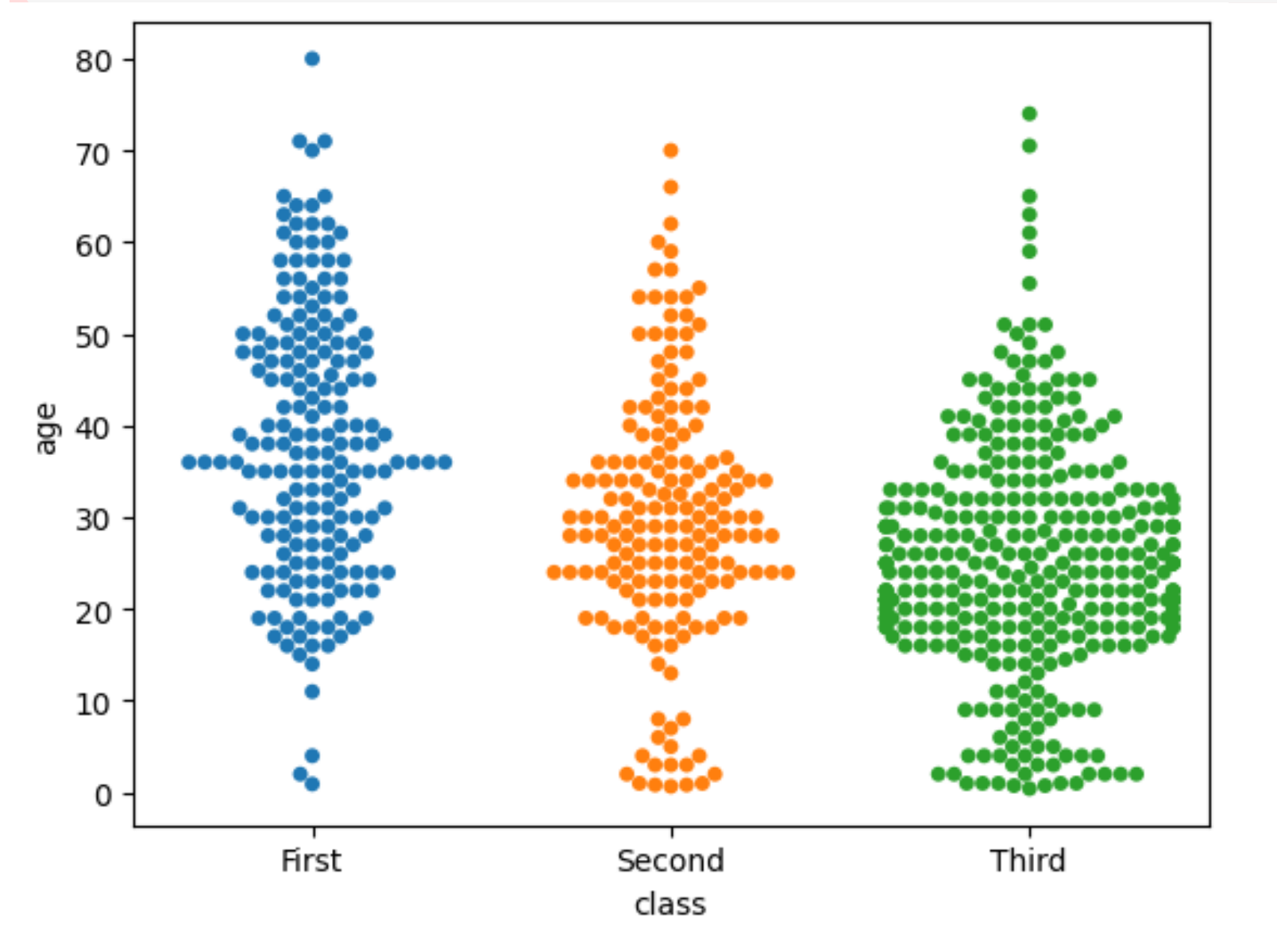




In […
```
sns.swarmplot(x='class', y='age', data=titanic, hue='class')
```

Out[…
```
<Axes: xlabel='class', ylabel='age'>
```

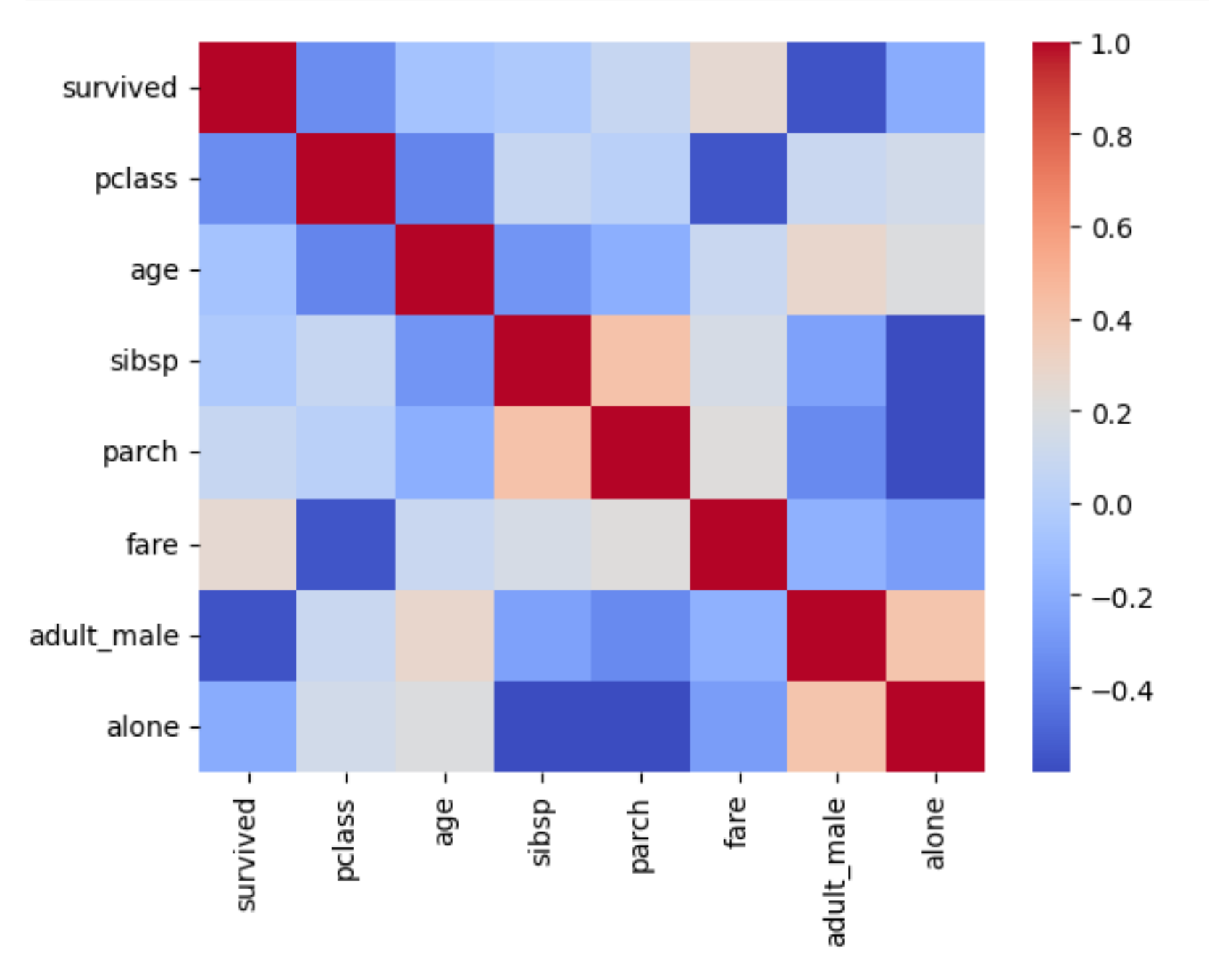
```
/Users/nick/Library/Python/3.9/lib/python/site-packages/seaborn/categorical.py:3399: UserWarning: 15.2% of the points cannot be placed; you may want to decrease the size of the markers or use stripplot.
  warnings.warn(msg, UserWarning)
```



sns.heatmap()

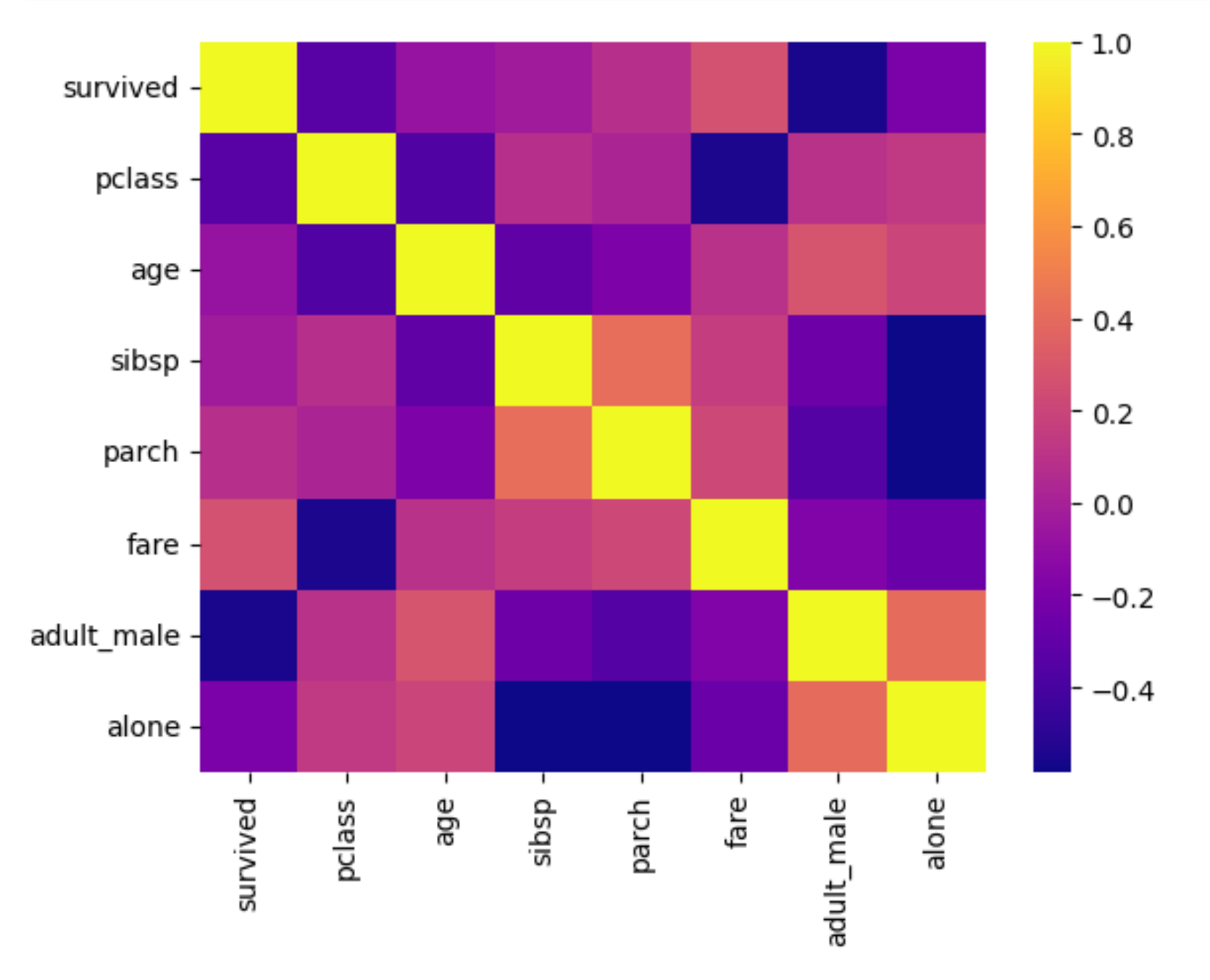
In […
```
sns.heatmap(titanic.corr(numeric_only=True),cmap='coolwarm')
```

Out[…
```
<Axes: >
```



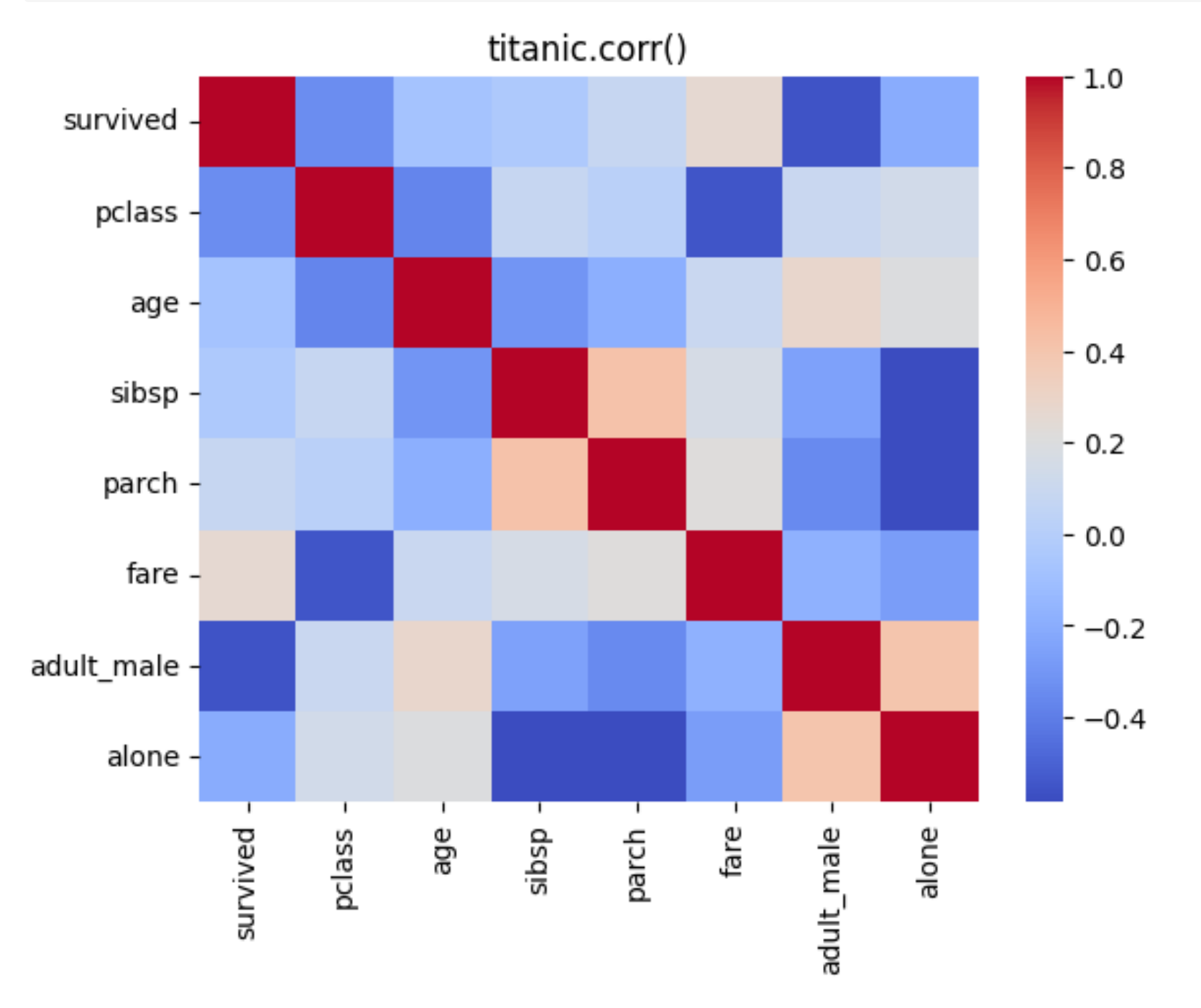
In [7…
```
sns.heatmap(titanic.corr(numeric_only=True),cmap='plasma')
```

Out[7…
```
<Axes: >
```



In [7…
```
sns.heatmap(titanic.corr(numeric_only=True),cmap='coolwarm')
plt.title('titanic.corr()')
```

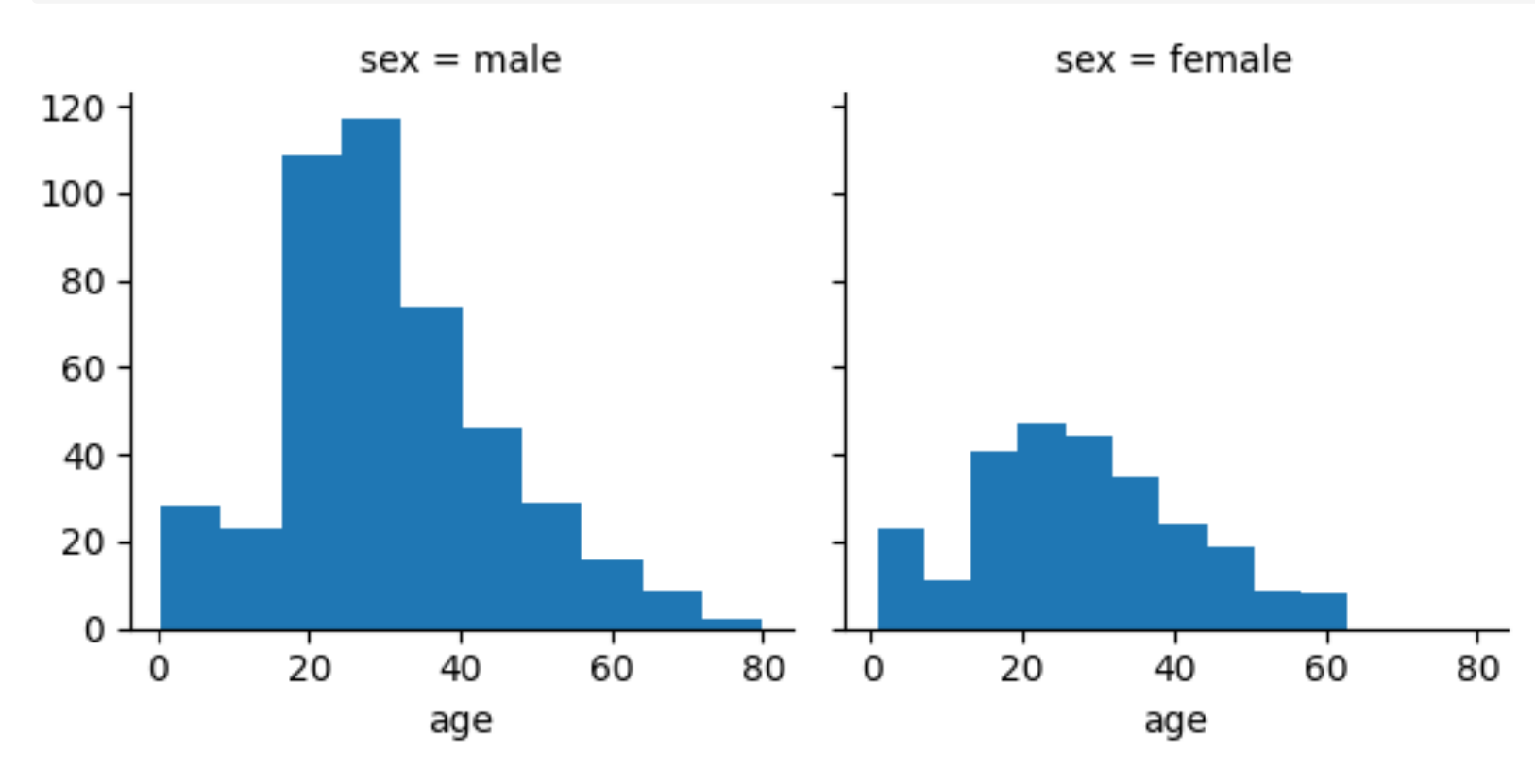
Out[7…
```
Text(0.5, 1.0, 'titanic.corr()')
```



sns.FacetGrid()

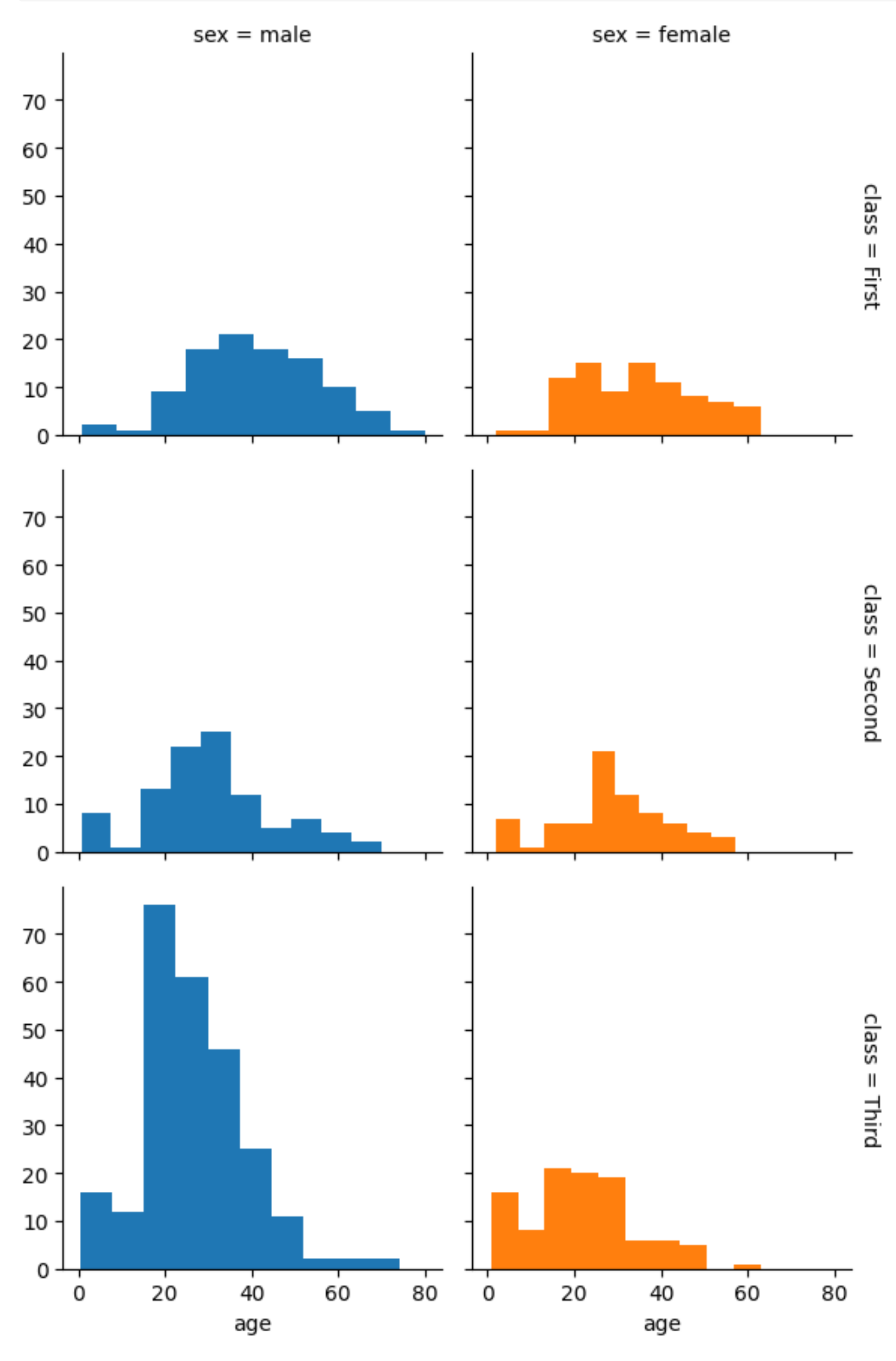
In [7…
```
g = sns.FacetGrid(data=titanic, col='sex')
g.map(plt.hist,'age')
```

Out[7…
```
<seaborn.axisgrid.FacetGrid at 0x1566e06a0>
```



In […
```
g = sns.FacetGrid(data=titanic,col='sex', row='class', hue='sex', margin_titles=True)
g.map(plt.hist,'age')
```

Out[…
```
<seaborn.axisgrid.FacetGrid at 0x1567df5e0>
```



This Jupyter Notebook contains exercises for you to extend your introduction to the basics with Python Seaborn. Attempt the following exercises, which slowly build in complexity. If you get stuck, check back to the Python lecture recording on Seaborn here or view the W3Schools page on Python Seaborn, which includes examples, exercises and quizzes to help your understanding.

Seaborn Exercises (Jose Portila)

Time to practice your new seaborn skills! Try to recreate the plots below (don't worry about color schemes, just the plot itself.

Exercises

** Recreate the plots below using the titanic dataframe. There are very few hints since most of the plots can be done with just one or two lines of code and a hint would basically give away the solution. Keep careful attention to the x and y labels for hints.**

** *Note! In order to not lose the plot image, make sure you don't code in the cell that is directly above the plot, there is an extra cell above that one which won't overwrite that plot!* **

The Data

We will be working with a famous titanic data set for these exercises. Later on in the Machine Learning section of the course, we will revisit this data, and use it to predict survival rates of passengers. For now, we'll just focus on the visualization of the data with seaborn:

In [1…
```
import seaborn as sns
import matplotlib.pyplot as plt
%matplotlib inline
```

In [2…
```
sns.set_style('whitegrid')
```

In [3…
```
titanic = sns.load_dataset('titanic')
```

In [5…
```
titanic.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 891 entries, 0 to 890
Data columns (total 15 columns):
 #   Column       Non-Null Count  Dtype   
---  ------       --------------  -----   
 0   survived     891 non-null    int64   
 1   pclass       891 non-null    int64   
 2   sex          891 non-null    object  
 3   age          714 non-null    float64 
 4   sibsp        891 non-null    int64   
 5   parch        891 non-null    int64   
 6   fare         891 non-null    float64 
 7   embarked     889 non-null    object  
 8   class        891 non-null    category
 9   who          891 non-null    object  
 10  adult_male   891 non-null    bool    
 11  deck         203 non-null    category
 12  embark_town  889 non-null    object  
 13  alive        891 non-null    object  
 14  alone        891 non-null    bool    
dtypes: bool(2), category(2), float64(2), int64(4), object(5)
memory usage: 80.7+ KB
```

In [4…
```
titanic.head()
```

Out[4…

	survived	pclass	sex	age	sibsp	parch	fare	embarked	class	who	adult_male	deck	embark_town	alive	alone
0	0	3	male	22.0	1	0	7.2500	S	Third	man	True	NaN	Southampton	no	False
1	1	1	female	38.0	1	0	71.2833	C	First	woman	False	C	Cherbourg	yes	False
2	1	3	female	26.0	0	0	7.9250	S	Third	woman	False	NaN	Southampton	yes	True
3	1	1	female	35.0	1	0	53.1000	S	First	woman	False	C	Southampton	yes	False
4	0	3	male	35.0	0	0	8.0500	S	Third	man	True	NaN	Southampton	no	True

Exercises

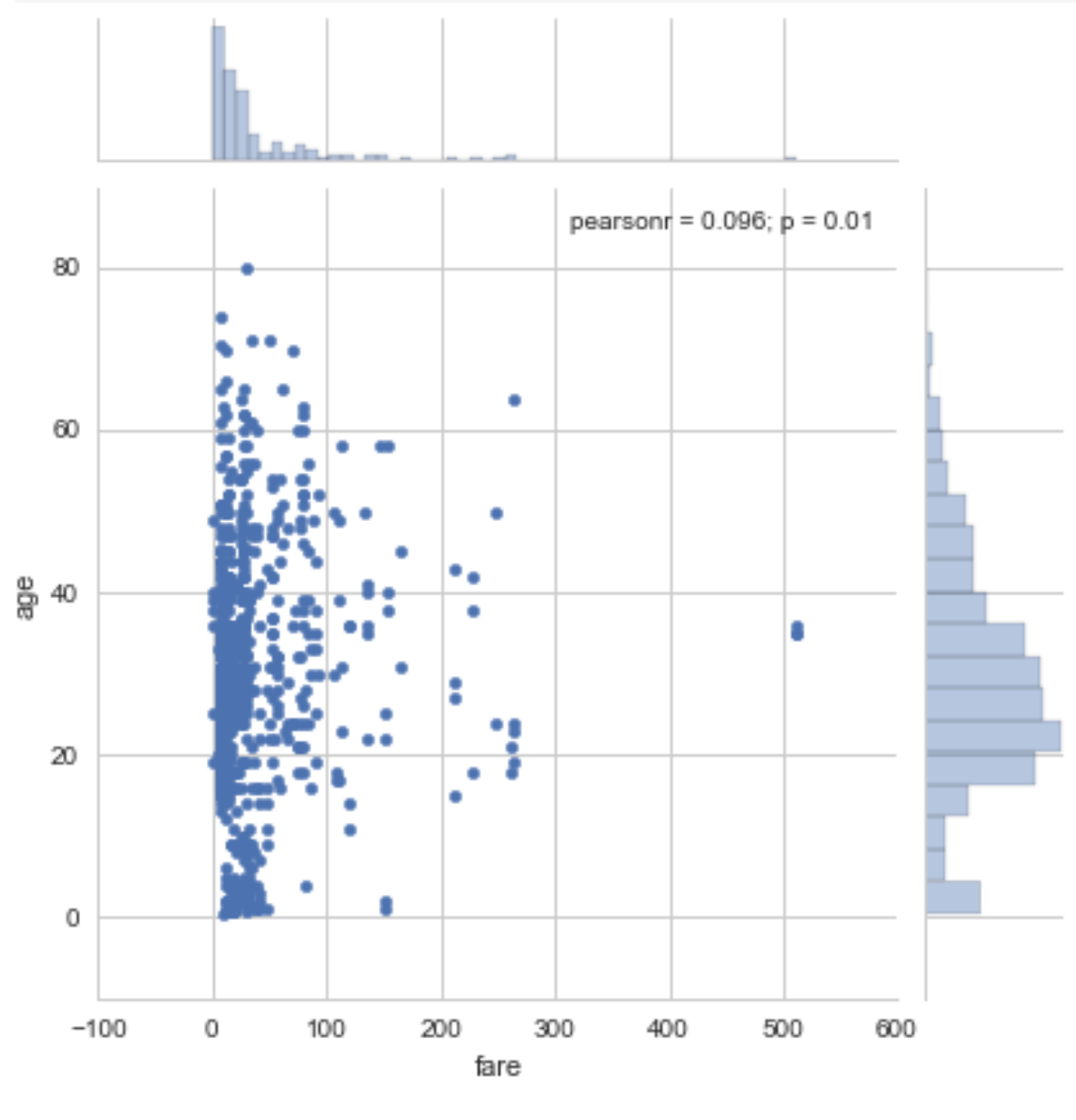
** Recreate the plots below using the titanic dataframe. There are very few hints since most of the plots can be done with just one or two lines of code and a hint would basically give away the solution. Keep careful attention to the x and y labels for hints.**

** *Note! In order to not lose the plot image, make sure you don't code in the cell that is directly above the plot, there is an extra cell above that one which won't overwrite that plot!* **

In […
```
# CODE HERE
# REPLICATE EXERCISE PLOT IMAGE BELOW
# BE CAREFUL NOT TO OVERWRITE CELL BELOW
# THAT WOULD REMOVE THE EXERCISE PLOT IMAGE!
```

In […

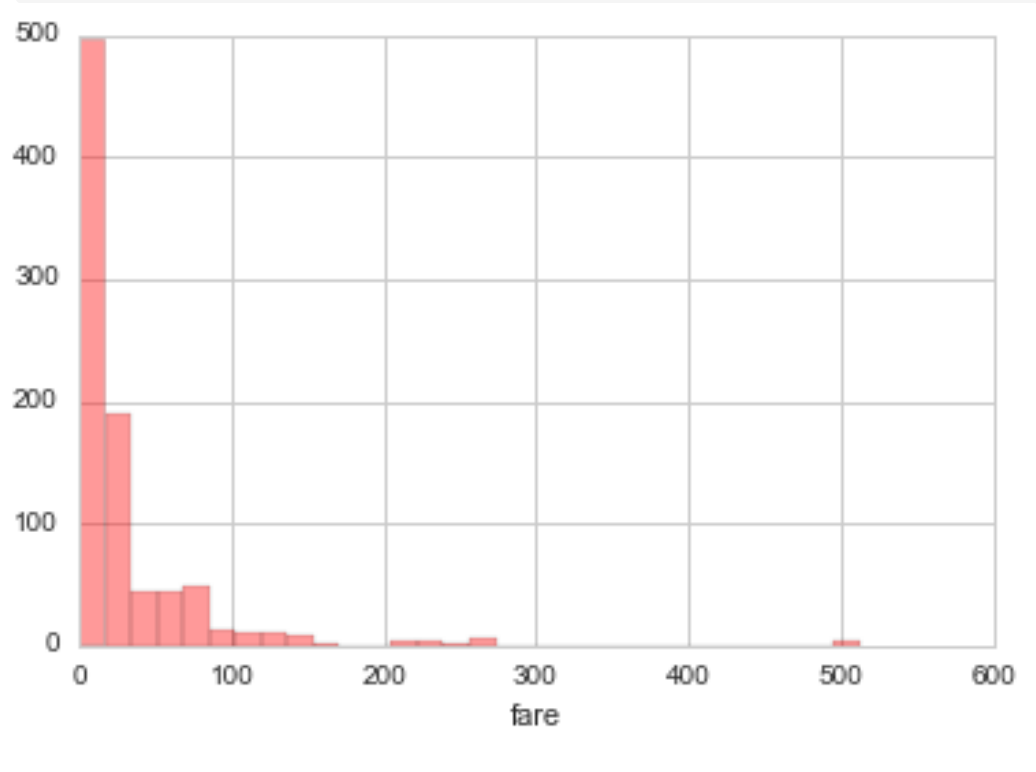
```
<seaborn.axisgrid.JointGrid at 0x11d0389e8>
```



In […
```
# CODE HERE
# REPLICATE EXERCISE PLOT IMAGE BELOW
# BE CAREFUL NOT TO OVERWRITE CELL BELOW
# THAT WOULD REMOVE THE EXERCISE PLOT IMAGE!
```

In […

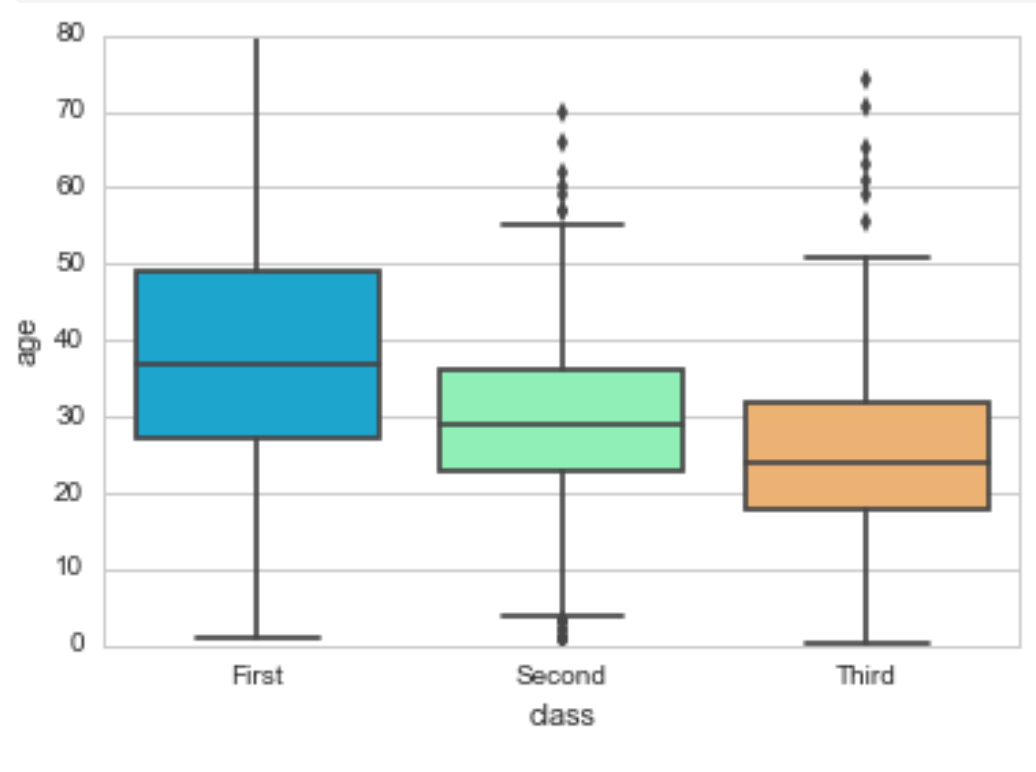
```
<matplotlib.axes._subplots.AxesSubplot at 0x11fc5ca90>
```



In […
```
# CODE HERE
# REPLICATE EXERCISE PLOT IMAGE BELOW
# BE CAREFUL NOT TO OVERWRITE CELL BELOW
# THAT WOULD REMOVE THE EXERCISE PLOT IMAGE!
```

In […

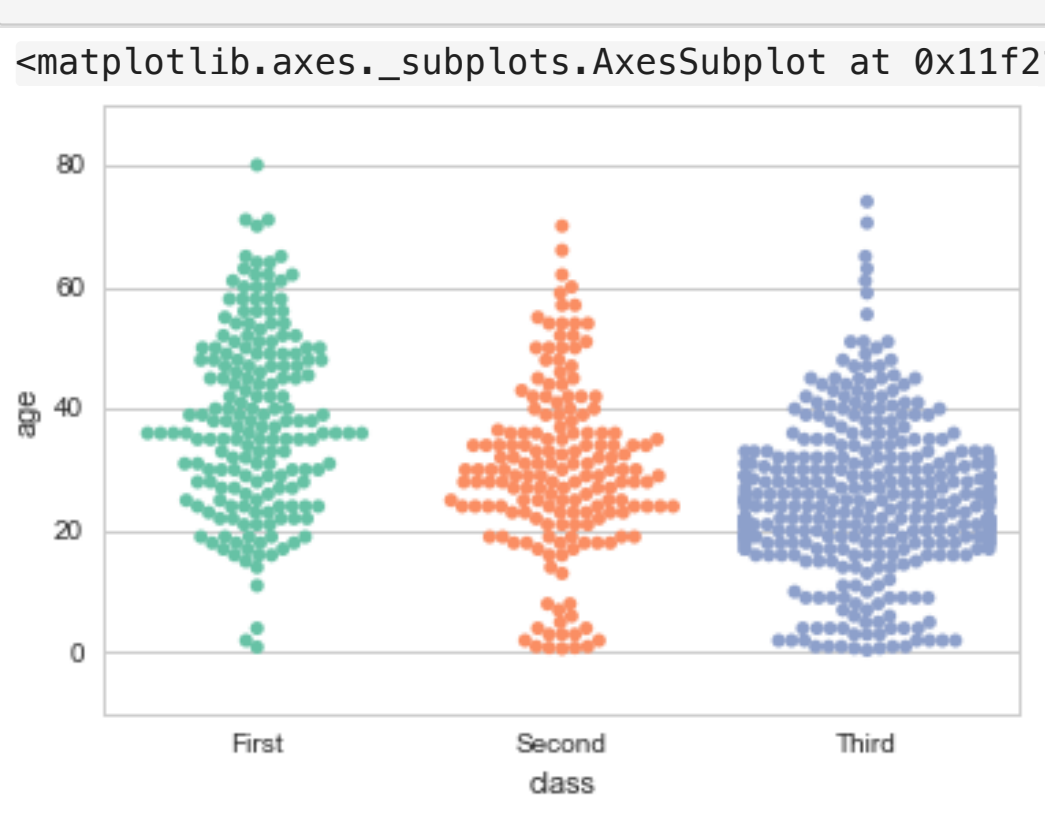
```
<matplotlib.axes._subplots.AxesSubplot at 0x11f23da90>
```



In […
```
# CODE HERE
# REPLICATE EXERCISE PLOT IMAGE BELOW
# BE CAREFUL NOT TO OVERWRITE CELL BELOW
# THAT WOULD REMOVE THE EXERCISE PLOT IMAGE!
```

In […

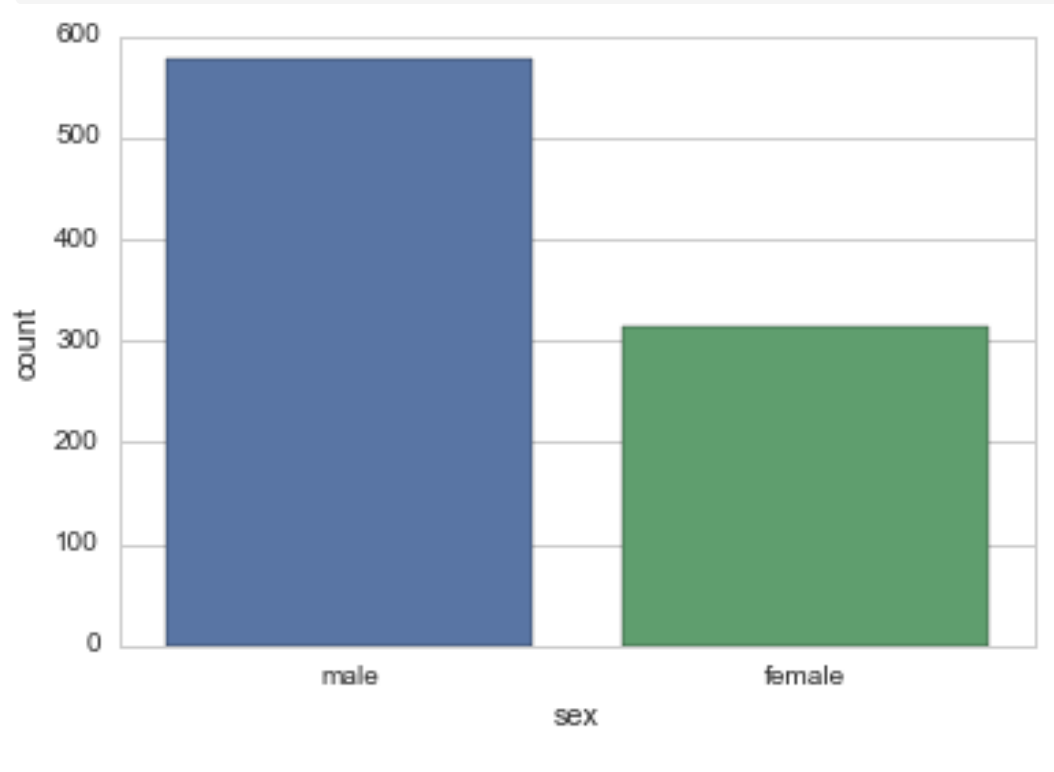
```
<matplotlib.axes._subplots.AxesSubplot at 0x11f215320>
```



In […
```
# CODE HERE
# REPLICATE EXERCISE PLOT IMAGE BELOW
# BE CAREFUL NOT TO OVERWRITE CELL BELOW
# THAT WOULD REMOVE THE EXERCISE PLOT IMAGE!
```

In […

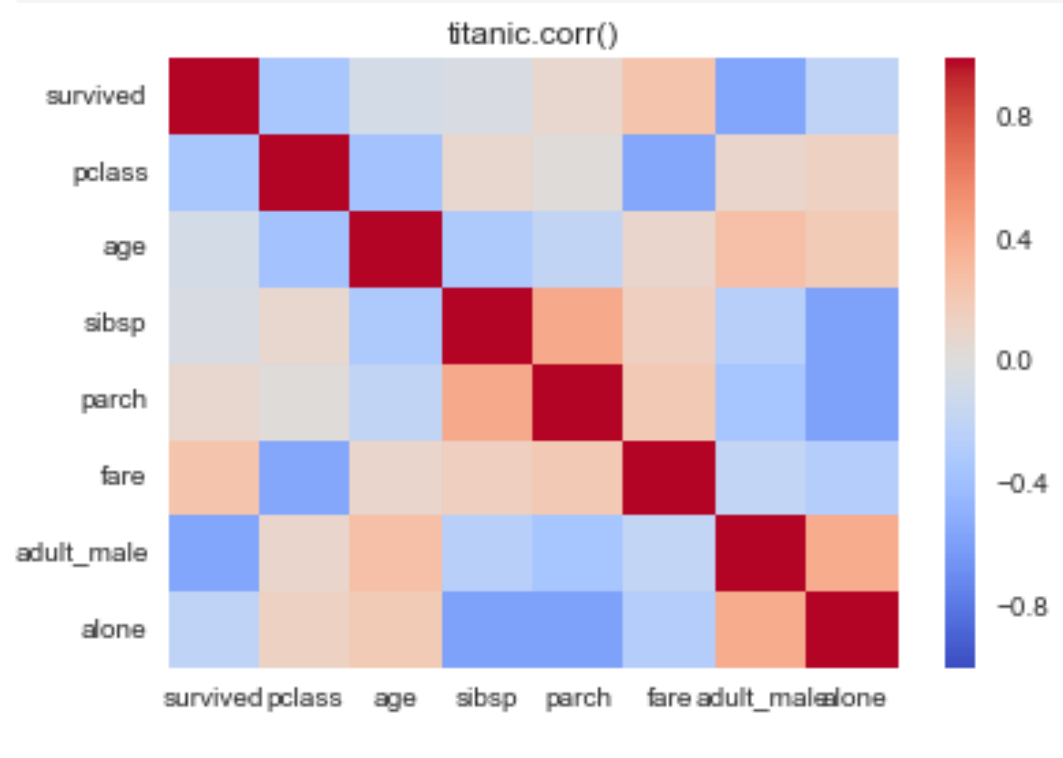
```
<matplotlib.axes._subplots.AxesSubplot at 0x11f207ef0>
```



In […
```
# CODE HERE
# REPLICATE EXERCISE PLOT IMAGE BELOW
# BE CAREFUL NOT TO OVERWRITE CELL BELOW
# THAT WOULD REMOVE THE EXERCISE PLOT IMAGE!
```

In […

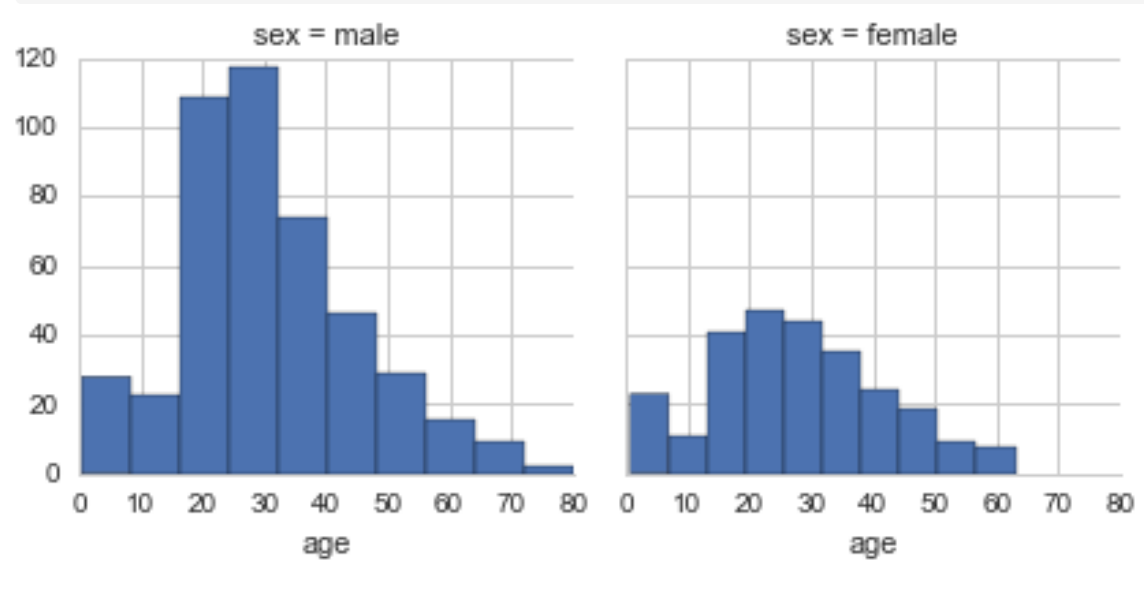
```
<matplotlib.text.Text at 0x11d72da58>
```



In […
```
# CODE HERE
# REPLICATE EXERCISE PLOT IMAGE BELOW
# BE CAREFUL NOT TO OVERWRITE CELL BELOW
# THAT WOULD REMOVE THE EXERCISE PLOT IMAGE!
```

In […

```
<seaborn.axisgrid.FacetGrid at 0x11d81c240>
```



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